

Compartmental Epidemiological Models with Infection-Driven Viral Mutation

Abstract

We propose and study a compartmental epidemiology model that accounts for both immune escape (via viral mutation) and waning immunity through a cross-immune group. The model, referred to as SIRCm, is a modification of the SIRC model in which both effects are driven by a mutation feedback tied the current level of infection. We study two variants of this feedback, one driven by the transmission term SI and the other by the infected fraction I alone. For both variants, we prove the existence of an endemic equilibrium. Unlike the SIRC model, we show that the SIRCm endemic equilibrium is unstable in certain parameter regimes. This leads to novel behavior, even in the absence of seasonal forcing.

1 Background and Motivation

The classical SIR (Susceptible–Infected–Recovered) model and its variants have proven surprisingly effective at describing the spread of infectious diseases at the population level. A central assumption in the basic SIR framework is that recovered individuals acquire permanent immunity, so that each individual is infected at most once. In practice, two distinct mechanisms can invalidate this assumption: (1) waning of immune memory over time, and (2) viral mutation resulting in immune escape, whereby the pathogen evolves to circumvent the defences of previously infected hosts.

The first mechanism—immunity waning—is well represented in the literature through SIRS-type models, in which recovered individuals return to the susceptible class at some fixed rate [10, 4]. The second mechanism—mutation-driven immune escape—has received considerably less attention as a modelling target in its own right. Of particular importance is the observation that mutations do not occur at a fixed rate in time; rather, mutation opportunities arise proportionally to the number of active infections, since each replication event inside a host is a potential mutation event [1, 3]. This coupling between infection prevalence and mutation supply is fundamental: it implies that the evolutionary trajectory of a pathogen is not independent of epidemic dynamics, but is instead shaped by them.

The COVID-19 pandemic has made this interaction especially visible. Successive variants of concern—Alpha, Delta, Omicron, and its subvariants—demonstrated that immune escape, rather than waning immunity alone, was the dominant driver of renewed epidemic waves [13, 11]. The pandemic also generated an unprecedented volume of genomic sequence data, providing new opportunities to calibrate and validate mathematical models of viral evolution at the population level [9].

In this work we propose and investigate an extension of the SIRC model [4, 7], in which additional terms represent the effect of viral immune escape. Because this model incorporates the effect viral *mutation* (distinct from that of the host’s waning immunity) we refer to it as SIRCm.

The paper is organized as follows. In Section 2, we describe the model and some of its basic properties. In Section 3, we characterize two types of equilibria (disease-free and endemic). In Section 4 we carry out a local stability and bifurcation analysis, illustrating the results. Section 5 discusses the biological implications of these findings.

As we show in Section 4, the model interpolates between two frameworks: in the absence of mutation feedback it reduces to the SIRC model of Casagrandi et al. [4], while in the limit of strong mutation feedback it reduces to the classical SIS model, in which recovery confers no immunity.

2 Model formulation

We propose and study the following system of ordinary differential equations (ODEs), henceforth referred to as the *SIRCm model*:

$$\dot{S} = \mu(1 - S) - \beta SI + (1 + \varepsilon_2 g(x))\gamma C, \quad (1a)$$

$$\dot{I} = \beta SI + \sigma \beta CI - (\mu + \alpha)I, \quad (1b)$$

$$\dot{R} = (1 - \sigma)\beta CI + \alpha I - \mu R - (1 + \varepsilon_1 g(x))\delta R, \quad (1c)$$

$$\dot{C} = (1 + \varepsilon_1 g(x))\delta R - \beta CI - \mu C - (1 + \varepsilon_2 g(x))\gamma C, \quad (1d)$$

with state variables $S, I, R, C \in [0, +\infty)$ and parameters

$$\mu, \beta, \alpha, \delta, \gamma > 0, \quad \sigma \in [0, 1], \quad \varepsilon_1, \varepsilon_2 \geq 0.$$

As usual, the overdot symbol denotes the time derivative. The solution vector of (1)—with suitable initial conditions—will be denoted by

$$x(t) := (S(t), I(t), R(t), C(t)).$$

The dependent variables represent the same compartments as in the SIRC model: *susceptible* (S), *infectious* (I), *recovered* (R), and *cross-immune* (C). Each represents a fraction of the total population, and takes values in $[0, 1]$. Most of the parameters are also identical: μ is the *natural birth* (or mortality) *rate*, assumed equal across compartments so that the total population is conserved; $1/\mu$ is the average host lifetime. The parameter β is the *transmission rate*, governing new infections at rate βSI , and α is the *recovery rate* from active infection, with $1/\alpha$ being the average infectious period. The parameter δ is the rate at which *complete immunity wanes* into the cross-immune class C ($R \rightarrow C$), and γ is the rate at which *cross-immunity* itself *wanes* into full susceptibility ($C \rightarrow S$); $1/\delta$ and $1/\gamma$ are the corresponding average times spent in each immune state before transitioning onward. The parameter σ is the fraction of C – I contacts that result in *renewed active infection* rather than successful clearance. Finally, ε_1 and ε_2 are the *mutation feedback parameters*, scaling how strongly the current level of active infection amplifies immune-escape δ and waning γ , respectively, via the terms $(1 + \varepsilon_1 g(x))$ and $(1 + \varepsilon_2 g(x))$ multiplying each. Setting $\varepsilon_1 = \varepsilon_2 = 0$ recovers the SIRC model of Casagrandi et al. [4]. Figure 1 illustrates the compartments and transition rates of the SIRCm model. Throughout the paper, we explore two formulations of g , differing in how the mutation feedback is driven by the state of the system:

- the *transmission-driven variant*, in which mutation opportunities arise at transmission events, given by

$$g(x) = SI;$$

- the *prevalence-driven variant*, in which mutation opportunities arise from the currently infected population, independent of ongoing transmission, given by

$$g(x) = I.$$

These variations are compared directly in Section 4.

The *basic reproduction number*, denoted by $\mathcal{R}_0$, is the expected number of secondary infections generated by a single infectious individual in an otherwise fully susceptible population. For the SIRCm model (1), we have (see Proposition 2.4)

$$\mathcal{R}_0 := \frac{\beta}{\mu + \alpha}. \quad (2)$$

Realistic ranges for the parameters are shared with those stated by Casagrandi et al. in [4, Table 1] for Influenza A. Unless otherwise noted, the base parameter values used throughout the numerical results of this paper follow those of [4, Fig. 2]: $\mu = 0.02$, $\alpha = 365/3$, $\delta = 1.6^{-1}$, $\sigma = 0.07874$, $\gamma = 0.35$, with β_0 and the mutation feedback parameters $\varepsilon_1, \varepsilon_2$ varied as indicated in each figure.

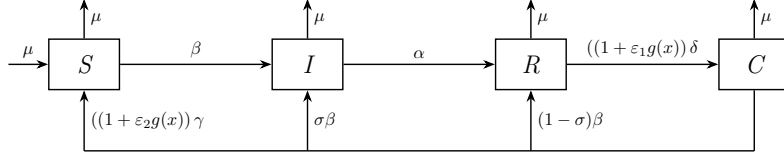


Figure 1: Flow diagram of the SIRCm model.

2.1 Preliminaries

In this section, we introduce the state space and list some of its elementary properties.

The state space is defined as

$$\Omega := \{(S, I, R, C) \in [0, 1]^4 : S + I + R + C = 1\}$$

equipped with the standard Euclidean metric.

Definition 2.1. Let g be a continuously differentiable function on Ω , satisfying:

- (a) $g(x)$ depends exclusively on the state variables S and I
- (b) $g \geq 0$ on Ω
- (c) $I = 0 \iff g = 0$

Then the following propositions hold.

Proposition 2.2. $\Omega \subset \mathbb{R}^4$, the standard simplex, is non-empty, compact, and convex.

Proposition 2.3. Ω is forward invariant, that is, for any initial condition $x(0) \in \Omega$, the unique solution of (1) satisfies $x(t) \in \Omega$ for all $t \geq 0$.

Proof. Conservation. Let $P := S + I + R + C$. Summing the four equations of (1) yields $\dot{P} = \mu(1 - P)$. This solution of this linear scalar ODE is $P(t) = 1 + (P(0) - 1)e^{-\mu t}$. In particular, $P(0) = 1 \implies P(t) = 1$ for all $t \geq 0$.

Non-negativity. We verify that the vector field is quasi-positive on $[0, +\infty)^4$, i.e., $\dot{x}_i \geq 0$ whenever $x_i = 0$ ($1 \leq i \leq 4$) and the remaining coordinates are non-negative.

- For I : $\dot{I} = I[\beta S + \sigma\beta C - (\mu + \alpha)]$ is linear in I , with solution

$$I(t) = I(0) \exp\left(\int_0^t [\beta S(u) + \sigma\beta C(u) - (\mu + \alpha)] du\right).$$

Since the exponential factor is strictly positive, $I(0) \geq 0 \implies I(t) \geq 0$ for all $t \geq 0$.

- For S : $\dot{S}|_{S=0} = \mu + (1 + \varepsilon_2 g(x))\gamma C > 0$, since $g(x) \geq 0$.
- For R : $\dot{R}|_{R=0} = (1 - \sigma)\beta C I + \alpha I \geq 0$.
- For C : $\dot{C}|_{C=0} = (1 + \varepsilon_1 g(x))\delta R \geq 0$, since $g(x) \geq 0$.

Nagumo's theorem [2] then implies $x(t) \geq 0$, and—together with the conservation property— $x(t) \in \Omega$ for all $t \geq 0$ for which the solution of (1) is defined.

The right-hand sides of (1) are polynomials in (S, I, R, C) , hence they are locally Lipschitz and continuous functions, guaranteeing local existence and uniqueness of the corresponding initial-value problem. Finally, from the extension theorem for ODEs, we get that the solution $x(t)$ is defined for *all* $t \geq 0$, since the solution does not leave the compact set Ω . $\square$

Similarly to SIRC, we point out the existence of a disease-free equilibrium on the boundary of Ω . Its stability properties depend on the basic reproduction number.

Proposition 2.4. *SIRCM (1) has a unique disease-free equilibrium (DFE) on $\Omega_0 := \{x \in \Omega : I = 0\}$, $\overset{\circ}{x} := (1, 0, 0, 0)$. Setting $\mathcal{R}_0 := \beta/(\mu + \alpha)$, $\overset{\circ}{x}$ is hyperbolic whenever $\mathcal{R}_0 \neq 1$, locally asymptotically stable when $\mathcal{R}_0 < 1$, and unstable when $\mathcal{R}_0 > 1$. The eigenvalues of $J(\overset{\circ}{x})$ are:*

$$\{-\alpha + \beta - \mu, -\delta(1 + g(\overset{\circ}{x})\varepsilon_1) - \mu, -\gamma(1 + g(\overset{\circ}{x})\varepsilon_2) - \mu, -\mu\}.$$

3 The endemic equilibrium

In this section, we show the existence of a second equilibrium in the interior of Ω (denoted by $\text{int}(\Omega)$).

The following proposition records the fact that the solution map of the autonomous ODE (1) generates a continuous semiflow (since now its right-hand side is sufficiently smooth).

Proposition 3.1. *Define $\Phi(t, x(0)) := x(t)$, where $x(\cdot)$ is the solution of (1) with initial state $x(0)$. Then $\Phi : [0, +\infty) \times \Omega \rightarrow \Omega$ is a continuous semiflow on Ω , in the sense of [12, Definition 1.33].*

The following auxiliary statement uses the concept of the ω -limit set, $\omega(\cdot)$, in the sense of [12, Definition 2.7].

Proposition 3.2. *Set $\mathcal{A} := \omega(\Omega)$. Then $\mathcal{A}$ is a non-empty, compact, invariant subset of Ω that attracts every subset of Ω ; that is, $\mathcal{A}$ is a compact attractor of neighborhoods of compact sets in Ω , in the sense of [12, Definition 2.22].*

Proof. Since Ω is compact (Proposition 2.2) and forward invariant under Φ (Proposition 2.3), every trajectory starting in Ω will remain there, hence any sequence $\Phi(t_i, x_i)$ with $t_i \rightarrow \infty$ and $x_i \in \Omega$ has a convergent subsequence. Therefore, Φ is asymptotically compact on Ω in the sense of [12, Definition 2.9].

By the equivalence (1) $\iff$ (3) of [12, Proposition 2.10], $\mathcal{A}$ is non-empty, compact, and attracts Ω . Since Φ is also asymptotically compact on Ω , [12, Theorem 2.11] also gives that $\mathcal{A}$ is invariant.

Since $\mathcal{A}$ attracts Ω , it also attracts every subset of Ω , in particular, $\mathcal{A}$ attracts every compact set in Ω , and every neighborhood of such set. Together with the invariance and compactness of $\mathcal{A}$, it is a compact attractor of neighborhoods of compact sets by [12, Definition 2.22]. $\square$

We define the *persistence function* $\rho : \Omega \rightarrow [0, +\infty)$ by $\rho(S, I, R, C) := I$, and the associated *extinction set*

$$\Omega_0 := \rho^{-1}(0) = \{x \in \Omega : I = 0\}.$$

Since ρ is a coordinate projection, it is continuous and linear, hence concave in particular. These definitions and properties will be needed in the following section.

3.1 The existence of an endemic equilibrium via persistence theory

By eliminating S, R, C from the equilibrium equations of SIRCM (1), we get a polynomial equation $\mathcal{P}(I, \varepsilon_1, \varepsilon_2) = 0$, quartic in I ; the explicit form of $\mathcal{P}$ is given in Appendix A. A root of $\mathcal{P}(\cdot, \varepsilon_1, \varepsilon_2)$ in $(0, 1]$ would yield an endemic equilibrium provided that $(S, I, R, C) \in \Omega$. However, by using standard algebraic techniques, we were unable to prove the existence of such an equilibrium in the simplex Ω , even when we reduced the dimensionality of the state space to a 3-dimensional tetrahedron by eliminating one of the variables from the conservation property $S + I + R + C = 1$. These difficulties motivate the persistence-theoretic approach [12] of the remainder of this section.

Proposition 3.3. *Ω_0 is forward invariant under the SIRCM (1) flow.*

Proof. Since $\dot{I} = I[\beta S + \sigma\beta C - (\mu + \alpha)]$ is linear in I , $I(0) = 0$ implies $I(t) \equiv 0$ for all $t \geq 0$. Hence Ω_0 is forward invariant. $\square$

Proposition 3.4. *Every trajectory on Ω_0 converges to $\overset{\circ}{x}$. Consequently, $\{\overset{\circ}{x}\}$ is invariant, is the global compact attractor of the boundary semiflow $\Phi|_{\Omega_0}$, and is the unique compact invariant subset of Ω_0 . Moreover the single-set family $\{M_1\} = \{\overset{\circ}{x}\}$ is an acyclic covering of $\tilde{\Omega} := \bigcup_{x \in \Omega_0} \omega(x)$ in the sense of [12, Section 8.3].*

Proof. Since $I = 0 \implies g = 0$ (Definition 2.1), (1) reduces on Ω_0 to the linear triangular system

$$\dot{R} = -(\mu + \delta)R, \quad \dot{C} = \delta R - (\mu + \gamma)C, \quad \dot{S} = \mu(1 - S) + \gamma C,$$

with $S + R + C = 1$. Solving in order yields $R(t) \rightarrow 0$, then $C(t) \rightarrow 0$, then $S(t) \rightarrow 1$ by conservation. Hence every trajectory on Ω_0 converges to $\overset{\circ}{x}$.

Since $\overset{\circ}{x}$ is an equilibrium, $\{\overset{\circ}{x}\}$ is invariant. Convergence of every trajectory gives $\tilde{\Omega} = \{\overset{\circ}{x}\}$, so $\{\overset{\circ}{x}\}$ is the global compact attractor of $\Phi|_{\Omega_0}$, and it is the only compact invariant subset of Ω_0 .

In the setup of [12, Section 8.3], the acyclicity theorem takes as input the following set:

$$\tilde{\Omega} := \bigcup_{x \in \Omega_0} \omega(x),$$

which is the union of ω -limits of all trajectories that start on the extinction boundary. [12, Theorem 8.17] then asks for a finite family $\{M_1, \dots, M_k\}$ of invariant sets in Ω_0 whose union covers $\tilde{\Omega}$. Since every trajectory on Ω_0 converge to $\overset{\circ}{x}$, we have $\omega(x) = \{\overset{\circ}{x}\}$ for every $x \in \Omega_0$, hence $\tilde{\Omega} = \{\overset{\circ}{x}\}$. Taking $k = 1$ and $M_1 = \{\overset{\circ}{x}\}$, the covering condition $\tilde{\Omega} \subset \bigcup_{j=1}^k M_j$ reduces to the trivial identity $\{\overset{\circ}{x}\} \subset \{\overset{\circ}{x}\}$.

Acyclicity is immediate from the graph-theoretic criterion of [12, Definition 8.14]: the associated directed graph has a single vertex M_1 and no self-loop, hence no cycles. $\square$

Proposition 3.5. *Assume $\mathcal{R}_0 \neq 1$, see (2). Then $\{\overset{\circ}{x}\}$ is isolated in Ω , in the sense of [12, Definition 5.14], that is, there exists a neighborhood V of $\overset{\circ}{x}$ in Ω such that every compact invariant $K \subset V$ satisfies $K \subset \{\overset{\circ}{x}\}$.*

Proof. Since $\mathcal{R}_0 \neq 1$, every eigenvalue of $J(\overset{\circ}{x})$ has nonzero real part (Proposition 2.4), so $\overset{\circ}{x}$ is a hyperbolic equilibrium. By the Hartman–Grobman theorem, Φ is topologically conjugate near $\overset{\circ}{x}$ to the linear flow $\dot{y} = J(\overset{\circ}{x})y$ near the origin. Since none of its eigenvalues have zero real part, this linear flow has no compact invariant set in a small neighborhood of 0 other than $\{0\}$ itself. Topological conjugacy transfers this to Φ , so the corresponding neighborhood V of $\overset{\circ}{x}$ contains no compact invariant set other than $\{\overset{\circ}{x}\}$. $\square$

Proposition 3.6. *Assume $\mathcal{R}_0 > 1$. Then $\{\overset{\circ}{x}\}$ is weakly ρ -repelling, in the sense of Definition 8.15 of [12]: there is no $x \in \Omega$ with $\rho(x) = I > 0$ such that $\Phi(t, x) \rightarrow \overset{\circ}{x}$ as $t \rightarrow \infty$.*

Proof. By Proposition 2.4, the stable eigenspace E^s of $J(\overset{\circ}{x})$ (spanned by the eigenvectors of $-\mu$, $-\delta(1 + g(\overset{\circ}{x})\varepsilon_1) - \mu$, and $-\gamma(1 + g(\overset{\circ}{x})\varepsilon_2) - \mu$) equals $\{I = 0\}$, the tangent space of Ω_0 at $\overset{\circ}{x}$. By the stable manifold theorem, there is a unique locally invariant manifold through $\overset{\circ}{x}$ tangent to E^s , called the local stable manifold $W_{\text{loc}}^s(\overset{\circ}{x})$. Since Ω_0 is itself invariant (Proposition 3.3) and tangent to E^s at $\overset{\circ}{x}$, uniqueness forces $W_{\text{loc}}^s(\overset{\circ}{x}) = \Omega_0 \cap V$ for some neighborhood V of $\overset{\circ}{x}$.

If $x \in \Omega$ has $I(x) > 0$ and $\Phi(t, x) \rightarrow \overset{\circ}{x}$, the trajectory would enter $W_{\text{loc}}^s(\overset{\circ}{x}) \subset \Omega_0$, forcing $I = 0$ at some time. This would contradict $I(0) > 0 \implies I(t) > 0$ for all t (Proposition 2.3). No such x exists. $\square$

Proposition 3.7. *Assume $\mathcal{R}_0 > 1$. Then Φ is uniformly weakly ρ -persistent: there exists $\eta_0 > 0$ such that*

$$\limsup_{t \rightarrow \infty} I(t) > \eta_0 \quad \text{whenever } I(0) > 0.$$

Proof. The standing hypothesis (H) of Section 8.3 of [12] holds with $B = \Omega$: since Ω is compact and forward invariant (Proposition 2.3), every trajectory trivially approaches Ω , and $\Omega \cap \{\rho \leq c\}$ is compact for every $c > 0$.

Now we apply [12, Theorem 8.17] to the family $\{M_1\} = \{\overset{\circ}{x}\}$: M_1 is compact and invariant (Proposition 3.4), isolated in Ω (Proposition 3.5), weakly ρ -repelling (Proposition 3.6), and acyclic as a single set. The theorem yields uniform weak ρ -persistence of Φ . $\square$

By applying [12, Theorem 4.5] with $B = \Omega$, we get the following.

Corollary 3.8. *Assume $\mathcal{R}_0 > 1$. Then Φ is uniformly (strongly) ρ -persistent:*

$$\liminf_{t \rightarrow \infty} I(t) > \eta_0 \quad \text{whenever } I(0) > 0.$$

Remark 3.9. [12, Theorem 6.2], used in Proposition 3.11 to establish the existence of the endemic equilibrium, only requires uniform weak persistence, however, Corollary 3.8 strengthens the biological conclusion that every solution with any initial disease presence where $\mathcal{R}_0 > 1$ has infection level ultimately bounded below by η_0 .

Proposition 3.10. *Assume $\mathcal{R}_0 > 1$. The set*

$$A_1 := \overline{\bigcup_{x \in \Omega \setminus \Omega_0} \omega(x)},$$

where $\overline{(\cdot)}$ denotes the closure operation, is a non-empty, compact, invariant subset of $\Omega \setminus \Omega_0$, and uniformly ρ -positive (i.e., there exists $\eta_1 > 0$ such that $\rho(x) \geq \eta_1$ for all $x \in A_1$).

Proof. Apply [12, Theorem 5.7]: since Ω is compact (Proposition 2.2) and Φ is uniformly weakly ρ -persistent (Proposition 3.7), the set A_1 has the stated properties. $\square$

Proposition 3.11. *Assume the basic reproduction number satisfies $\mathcal{R}_0 > 1$. Then, for any $\varepsilon_1, \varepsilon_2 \geq 0$, SIRCm (1) has an endemic equilibrium: there exists $x^* \in \Omega$ with $\rho(x^*) = I^* > 0$ and $\Phi(t, x^*) = x^*$ for all $t \geq 0$.*

Proof. We verify the hypotheses of [12, Theorem 6.2]. The set Ω is a closed and convex subset of the Banach space $\mathbb{R}^4$ (Proposition 2.2); ρ is continuous and concave (3); Φ is a continuous semiflow (Proposition 3.1); Φ has a compact attractor $\mathcal{A}$ of neighborhoods of compact sets (Proposition 3.2), and the persistence attractor A_1 exists (Proposition 3.10); Φ_t is condensing for all $t > 0$, and Φ is uniformly weakly ρ -persistent (Proposition 3.7).

It remains to verify hypothesis (H1): no total trajectory ϕ with range in $\mathcal{A}$ has $\rho(\phi(-r)) > 0$, $\rho(\phi(0)) = 0$, and $\rho(\phi(s)) > 0$ for some $r, s \geq 0$. By forward invariance of Ω_0 (Proposition 3.3), $I = 0$ at any time forces $I \equiv 0$ thereafter, so no trajectory can leave the set $\{I > 0\}$ and return, then hypothesis (H1) holds.

By [12, Theorem 6.2], there exists $x^* \in \Omega$ with $\rho(x^*) = I^* > 0$ and $\Phi(t, x^*) = x^*$ for all $t \geq 0$. $\square$

Proposition 3.12. *The coordinates of the equilibrium x^* in Proposition 3.11 satisfy $S^*, R^*, C^* > 0$. Hence $x^* \in \text{int}(\Omega)$.*

Proof. From $\dot{R} = 0$ at x^* :

$$(1 - \sigma)\beta C^* I^* + \alpha I^* = [\mu + (1 + \varepsilon_1 g(x^*))\delta] R^*.$$

The left-hand side is $\geq \alpha I^* > 0$, and the right-hand side is proportional to R^* with strictly positive coefficient (since $g(x^*) \geq 0$); hence $R^* > 0$.

From $\dot{C} = 0$ at x^* :

$$(1 + \varepsilon_1 g(x^*))\delta R^* = [\beta I^* + \mu + (1 + \varepsilon_2 g(x^*))\gamma] C^*.$$

The left-hand side is strictly positive since $\delta, R^* > 0$, and the right-hand side is proportional to C^* with strictly positive coefficient; hence $C^* > 0$.

From $\dot{S} = 0$ at x^* :

$$\mu + (1 + \varepsilon_2 g(x^*))\gamma C^* = [\mu + \beta I^*] S^*.$$

The left-hand side is at least $\mu > 0$, and the right-hand side is proportional to S^* with strictly positive coefficient, hence $S^* > 0$.

Together with $I^* \geq \eta_0 > 0$ (Proposition 3.7), all four coordinates of x^* are strictly positive, so $x^* \in \text{int}(\Omega)$. $\square$

We emphasize that the disease-free equilibrium's existence and stability (Proposition 2.4) and the persistence-theoretic existence of the endemic equilibrium (Proposition 3.11) were established for general g satisfying Definition 2.1.

These results continue to hold if we generalize the expression $(1 + \varepsilon_i g(x))$ by any function $h(\varepsilon_i, x)$ such that h is a continuously differentiable function that satisfies the following:

- $h(\varepsilon_i, x)$ depends exclusively on the parameter ε_1 or ε_2 , and the state variables S and I
- $h \geq 1$
- $I = 0 \iff h = 1$

4 Stability and bifurcation analysis

In general, the mutation feedback in SIRCm (1) is governed by two independent parameters ε_1 and ε_2 , present as amplification factors of the immune-escape and waning terms

$$(1 + \varepsilon_1 g(x)) \quad \text{and} \quad (1 + \varepsilon_2 g(x))$$

The size of this correction depends on the product $\varepsilon_i g(x)$. If $|\varepsilon_1 g(x)| \ll 1$ and $|\varepsilon_2 g(x)| \ll 1$, the mutation-driven correction is negligible relative to the baseline waning and immune-escape rates, and the model reduces to SIRC. If instead $|\varepsilon_1 g(x)|$ and $|\varepsilon_2 g(x)|$ are of order 1 or larger, the mutation feedback is comparable or exceeds the baseline rates, and the dynamics can differ significantly from SIRC.

Since S and I vary by orders of magnitude across parameter regimes, we introduce normalized parameters to remove this dependence:

$$\varepsilon_1 = \frac{\tilde{\varepsilon}_1}{g(x_0^*)}, \quad \varepsilon_2 = \frac{\tilde{\varepsilon}_2}{g(x_0^*)}. \quad (3)$$

With this reparametrization, the transition between the SIRC-like and mutation-dominated regimes occurs for $\tilde{\varepsilon}_1, \tilde{\varepsilon}_2$ close to $(0, 0)$, making $(\tilde{\varepsilon}_1, \tilde{\varepsilon}_2)$ the natural coordinates for exploring this feedback numerically.

Recall from Section 2 the transmission-driven variant ($g(x) = SI$) and the prevalence-driven variant ($g(x) = I$). Substituting into (3) recovers, respectively,

$$\varepsilon_i = \frac{\tilde{\varepsilon}_i}{S_0^* I_0^*} \quad \text{and} \quad \varepsilon_i = \frac{\tilde{\varepsilon}_i}{I_0^*}.$$

The remainder of this section will explore the local stability of the endemic equilibrium $x^* \in \text{int}(\Omega)$ established in Section 3.

Remark 4.1 (Notation). *We write x_ε^* for the endemic equilibrium of SIRCm (1) at a given value of the feedback parameter(s). The disease-free equilibrium $\hat{x}$ is as in Proposition 2.4. We further write $x_\varepsilon^* = (S^*, I^*, R^*, C^*)^\top$, where the coordinates inherit the same implicit dependence on ε . Since I^* is the primary object of interest below, we write it explicitly as I_ε^* when its ε -dependence is the focus.*

Figure 2 shows the Endemic prevalence I^* of the transmission-driven variant as a function of the normalized feedback parameters $\tilde{\varepsilon}_1$ and $\tilde{\varepsilon}_2$. The plot reveals two regimes separated by a sharp transition: a low prevalence regime, where I^* remains close to its SIRC value, and a high prevalence regime, where I^* is near its theoretical ceiling once either feedback parameter grows large. The right panel isolates the diagonal $\tilde{\varepsilon}_1 = \tilde{\varepsilon}_2 = \tilde{\varepsilon}$ and compares it against the first three orders of the low feedback Taylor expansion of 4.1.1. The prevalence-driven variant exhibits the same qualitative two-regime structure and comparable Taylor-series accuracy (Figure 16, Appendix B).

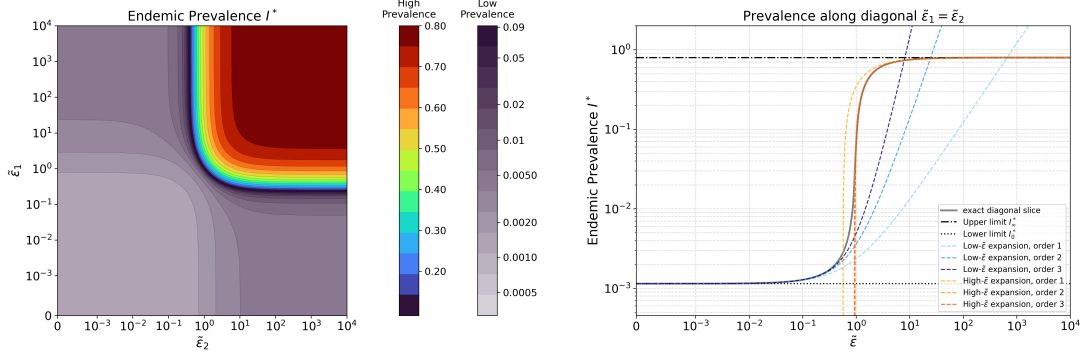


Figure 2: Endemic prevalence I^* of the transmission-driven variant of the SIRCm model. *Left*: two-dimensional plot of I^* as a function of ε_1 and ε_2 . *Right*: diagonal slice $\tilde{\varepsilon}_1 = \tilde{\varepsilon}_2$ together with the first three orders of the Taylor series approximations for low and high feedback regimes presented in Sections 4.1.1 and 4.3.2. The dotted line marks the lower limit I_0^* , the endemic prevalence of the SIRC model recovered as $\tilde{\varepsilon} \rightarrow 0$; the dash-dotted line marks the upper limit $I_\infty^* = 1 - \mathcal{R}_0^{-1}$, the theoretical ceiling on prevalence approached as $\tilde{\varepsilon} \rightarrow \infty$, corresponding to the SIS limit of Section 4.3. The color regimes transition at the inflection point of the endemic prevalence along the diagonal (where the derivative of I^* along $\varepsilon_1 = \varepsilon_2$ is maximized)

In order to analyze the stability of the endemic equilibrium, Figure 3 shows the Hopf bifurcation surface across $(\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \beta_0)$ for the prevalence-driven variant, with all other parameters fixed at the base values. An unstable region is present for sufficiently small $\tilde{\varepsilon}_1$. The diagonal $\varepsilon_1 = \varepsilon_2$, however, does not enter this unstable region for any $\beta_0 \in [200, 2000]$.

We adopt this single-parameter restriction $\varepsilon_1 = \varepsilon_2 =: \varepsilon$ for the remainder of this section, as it gives a natural one-parameter family suitable for direct comparison between the two variants.

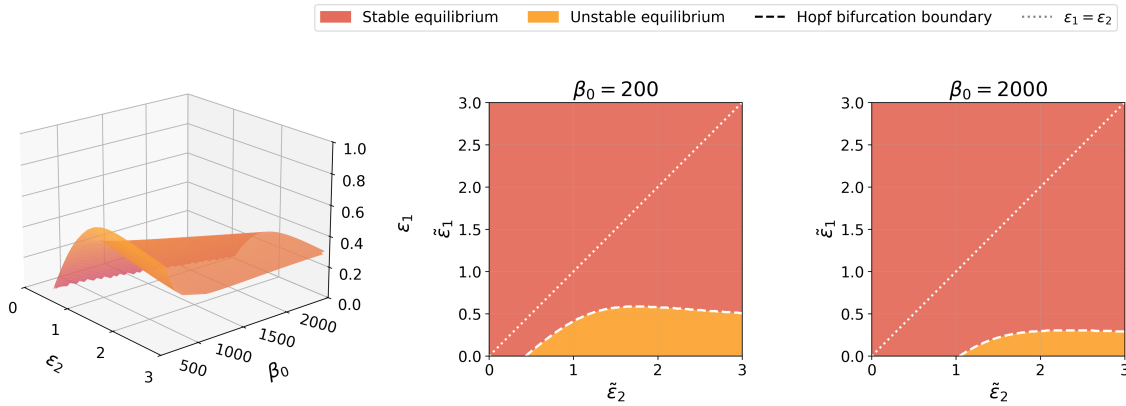


Figure 3: Hopf bifurcation surface across $(\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \beta_0)$ for the prevalence-driven variant (*left*), with slices at $\beta_0 = 200$ (*center*) and $\beta_0 = 2000$ (*right*), with all other parameters fixed

The Hopf bifurcation curves in Figures 3, 4, and 12 were computed in Julia using BifurcationKit.jl [14], via eigenvalue-crossing detection and pseudo-arclength continuation (PALC) of the resulting Hopf points.

Figure 4 shows the analogous Hopf bifurcation surface for the transmission-driven variant, with slices at $\beta_0 = 500$ and $\beta_0 = 900$. In contrast to the prevalence-driven case, the diagonal $\varepsilon_1 = \varepsilon_2$ does cross into the unstable region once β_0 is large enough.

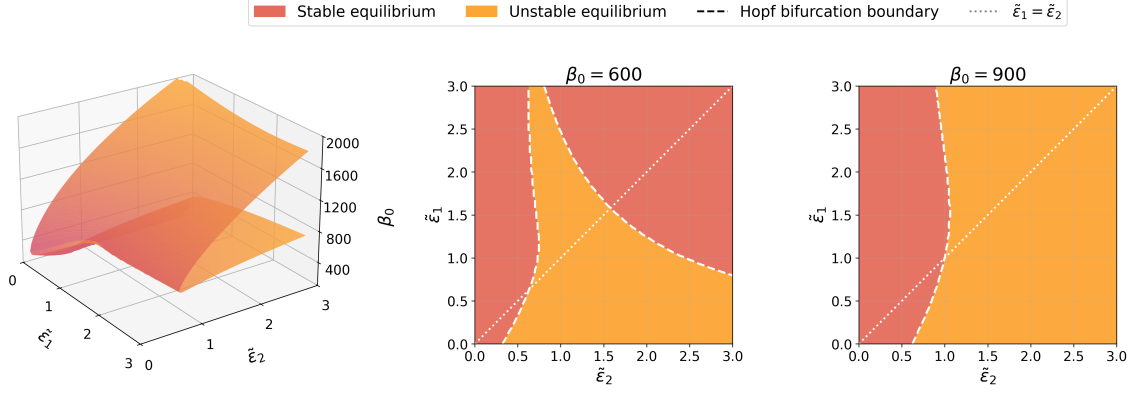


Figure 4: Hopf bifurcation surface across $(\tilde{\epsilon}_1, \tilde{\epsilon}_2, \beta_0)$ for the transmission-driven variant (*left*), with slices at $\beta_0 = 600$ (*center*) and $\beta_0 = 900$ (*right*), with all other parameters fixed

Because the transmission-driven variant exhibits a Hopf transition along the diagonal, we track the four eigenvalues of $J(x_\epsilon^*)$ as functions of $\tilde{\epsilon}$ at fixed β_0 . Figure 5 shows this sweep at $\beta_0 = 600$, where two eigenvalues remain negative throughout, while the remaining complex-conjugate pair crosses the imaginary axis twice, marking the two Hopf bifurcation points that bound the unstable region and partition $[0, 2]$ into the three regimes: the endemic equilibrium is locally asymptotically stable for smaller values of $\tilde{\epsilon}$, unstable for the middle region and stable again for larger values. The subsections that follow will examine each of these three regimes.

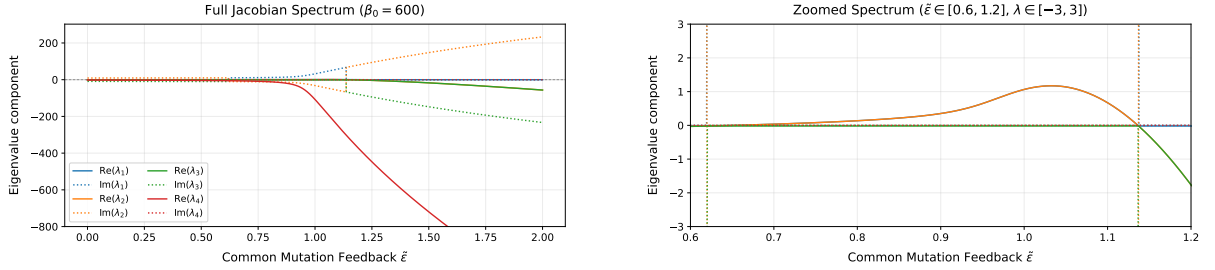


Figure 5: Eigenvalues of $J(x^*)$ as functions of $\tilde{\epsilon}$, at fixed $\beta_0 = 600$. *Left*: real and imaginary parts of all four eigenvalues over $\tilde{\epsilon} \in [0, 2]$. *Right*: zoom on the complex-conjugate pair whose real part changes sign, showing the two transversal crossings that mark the Hopf bifurcation points.

Figures 6 and 7 show the consequences of this loss and recovery of stability in the phase space of the system for the transmission-driven variant at one representative value of $\tilde{\epsilon}$ from each of the three regimes identified in Figure 5. At $\tilde{\epsilon} = 0.5$, trajectories spiral directly into the endemic equilibrium x_ϵ^* . At $\tilde{\epsilon} = 1.5$, inside the bifurcation region, the equilibrium is unstable and trajectories are instead attracted to a closed orbit encircling it, producing sustained oscillation in prevalence. At $\tilde{\epsilon} = 1.7$, stability is recovered and trajectories once again spiral into the equilibrium.

This sustained oscillation is notable because it arises without any seasonal forcing in the model; the periodicity is generated purely by the mutation feedback mechanism, in contrast to the externally imposed seasonality typically invoked to explain recurrent epidemic waves.

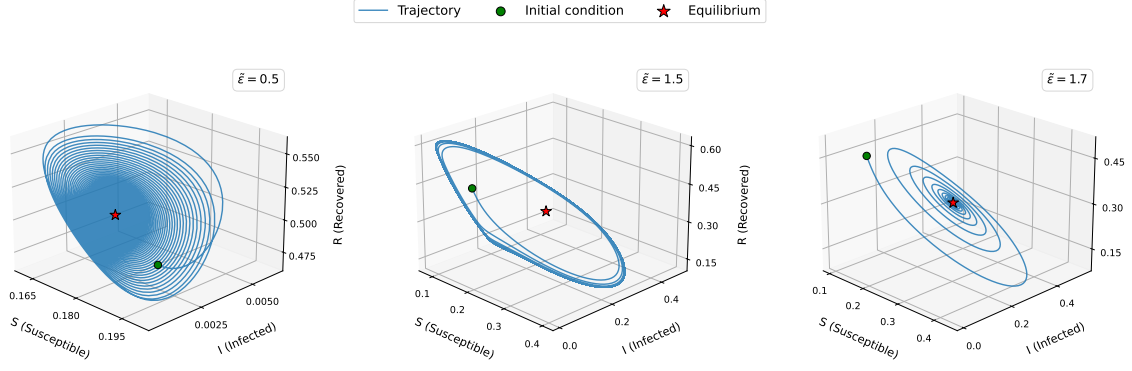


Figure 6: Trajectories in (S, I, R) space for three representative values of $\tilde{\varepsilon}$, spanning the weak, moderate and strong feedback regimes.

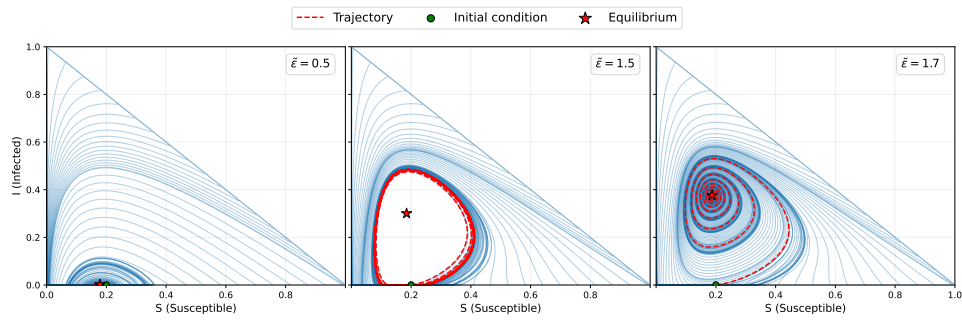


Figure 7: Phase portraits for the same three values of $\tilde{\varepsilon}$ as Figure 6, projected onto the (S, I) plane.

For the prevalence-driven variant, no comparable transition occurs along the diagonal. Trajectories and phase portraits converge directly to the endemic equilibrium at every value of $\tilde{\varepsilon}$ tested, as shown in Figures 8 and 9 for three representative values of $\tilde{\varepsilon}$.

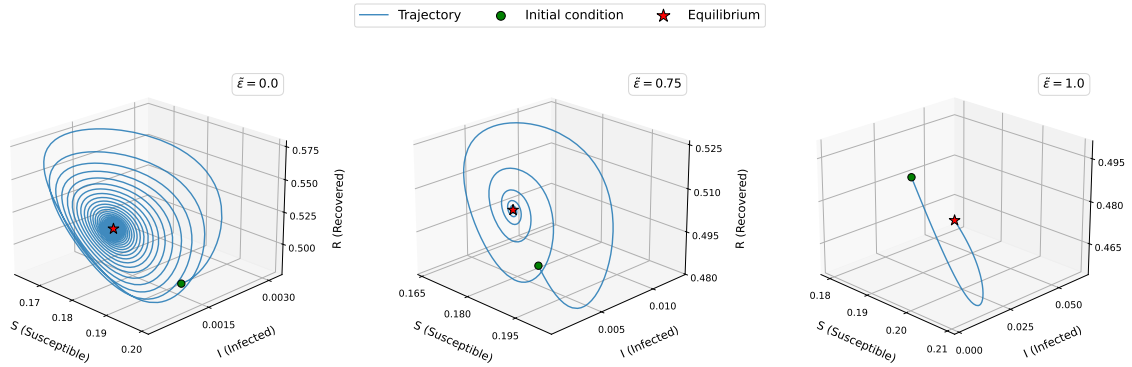


Figure 8: Trajectories in (S, I, R) space at $\tilde{\varepsilon} = 0, 0.75, 1$. All trajectories converge to the endemic equilibrium; trajectories for $\tilde{\varepsilon} > 1$ behave similarly to $\tilde{\varepsilon} = 1$.

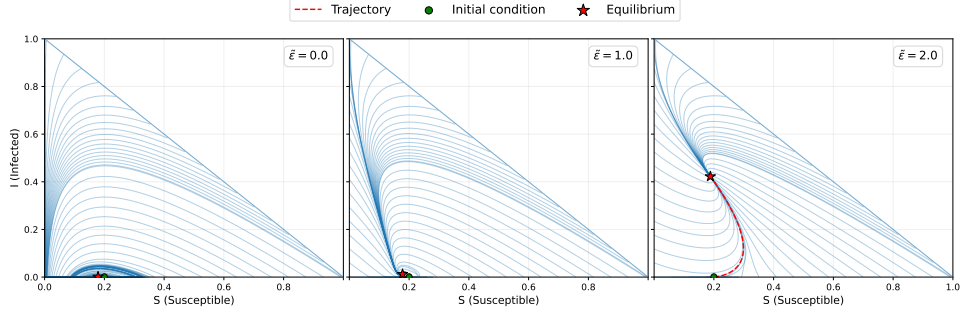


Figure 9: Phase portraits in the (S, I) plane at $\tilde{\varepsilon} = 0, 1, 2$.

The results presented in the remainder of this section can be summarized as follows:

- For weak and strong enough mutation effects, i.e. $\varepsilon \ll 1$ and $\varepsilon \gg 1$ respectively, we prove the uniqueness and asymptotic stability of the SIRCm endemic equilibrium, for both variants.
- For the transmission-driven variant, when mutation and waning effects balance each other, i.e. $\varepsilon = \mathcal{O}(1)$, we conjecture, supported by strong numerical evidence, that the endemic equilibrium is unique and I^* is a monotonically increasing function of ε . Moreover, we carry out a detailed bifurcation analysis, and show that the endemic equilibrium can become unstable as a consequence of mutation.
- For the prevalence-driven variant, by contrast, we find no such loss of stability along the diagonal $\varepsilon_1 = \varepsilon_2$ for any parameter values tested; the endemic equilibrium remains stable throughout.

4.1 Weak mutation feedback

[CM says: When I talk about neighborhoods, it's always in relative topologies wrt the semi-closed half-line for ε and the closed tetrahedron for x .]

Let us consider an endemic equilibrium x_ε^* of (1) (see Section 3.1). Since $I^* > 0$, from $\dot{I} = 0$ in (1b) we get

$$S^* = \mathcal{R}_0^{-1} - \sigma C^*. \quad (4)$$

Substituting into $I^* = 1 - S^* - R^* - C^*$ gives us

$$I^* = (1 - \mathcal{R}_0^{-1}) - R^* - (1 - \sigma)C^*, \quad (5)$$

which implies that $I_\infty^* = 1 - \mathcal{R}_0^{-1}$ is an upper bound on I^* for every ε , since $R^*, C^* \geq 0$ and $\sigma \in [0, 1]$.

Proposition 4.2. *Let $\mathcal{R}_0 > 1$ and let $g(x)$ satisfy Definition 2.1. Then, there exists an $\varepsilon^* > 0$ such that the SIRCm system (1) has a unique, asymptotically stable, endemic equilibrium for $\varepsilon \in [0, \varepsilon^*)$.*

Proof. Let us write the right-hand side of SIRCm (1) as

$$F(x, \varepsilon) = f(x) + \varepsilon g(x) \begin{bmatrix} \gamma C \\ 0 \\ -\delta R \\ -\gamma C \end{bmatrix}, \quad (6)$$

where $f(x)$ is the right-hand side of SIRC. Since $\mathcal{R}_0 > 1$, we know that SIRC has a unique, asymptotically stable endemic equilibrium x_0^* [4, Sec. 3], e.g.

$$F(x_0^*, 0) = 0, \quad \text{Re}(\lambda_{J_f}(x_0^*)) < 0. \quad (7)$$

In particular, we have that $\partial_x F(x_0^*, 0) = J_f(x_0^*)$ has non-zero determinant. Therefore, the Implicit Function Theorem guarantees the existence of $\varepsilon_{\text{IFT}} > 0$, a neighborhood of x_0^* , $U^* \subset \Omega$, and a unique continuously differentiable mapping $\psi^* : [0, \varepsilon] \rightarrow U^*$ such that $\psi^*(0) = x_0^*$ and

$$F(\psi^*(\varepsilon), \varepsilon) = 0, \quad (8)$$

for $\varepsilon \in [0, \varepsilon_{\text{IFT}})$.

Let us now show the existence of some $\underline{\varepsilon} > 0$, such that $\psi^*(\varepsilon)$ is the only endemic equilibrium in Ω (rather than just in U^*) for $0 < \varepsilon < \underline{\varepsilon}$. For the sake of contradiction, we assume the opposite, i.e. there is a sequence $\varepsilon_n \subset \mathbb{R}^+$ and $x_{\varepsilon_n}^* \subset \Omega$ of endemic equilibria such that $\varepsilon_n \rightarrow 0$ as $n \rightarrow +\infty$ and $x_{\varepsilon_n}^* \neq \psi^*(\varepsilon)$ for all $n \in \mathbb{N}$. Since Ω is compact, we can extract a convergent subsequence of $x_{\varepsilon_n}^*$, which we index by n for simplicity, with limit $x^* \in \Omega$. By the continuity of F , we have

$$F(x^*, 0) = \lim_{n \rightarrow \infty} F(x_{\varepsilon_n}, \varepsilon_n) = 0. \quad (9)$$

Therefore $x^* \in \{x_0^*, \hat{x}\}$, since SIRC only possesses two equilibria. From (4), we know that the continuous function $h(x) = S + \sigma C$ satisfies $h(x_{\varepsilon_n}^*) = \mathcal{R}_0^{-1}$ for every $n \in \mathbb{N}$. If $x^* = \hat{x}$, we get $1 = h(\hat{x}) = \lim_{n \rightarrow \infty} h(x_{\varepsilon_n}^*) = \mathcal{R}_0^{-1}$, which is a contradiction. Alternatively, if $x^* = x_0^*$, then there is a neighborhood $V \subset \Omega$ of x_0^* such that $x_{\varepsilon_n}^* \in V \cap U$ and $\varepsilon_n < \varepsilon_{\text{IFT}}$ for n large enough. However, $\psi^*(\varepsilon)$ is unique in $U \cap V$ for $\varepsilon < \varepsilon_{\text{IFT}}$, and we get a contradiction.

[DR says: @carlos here we should simplify the notation for λ]

Finally, given $0 < \varepsilon < \underline{\varepsilon}$, the local stability of the endemic equilibrium $\psi^*(\varepsilon)$ is determined by the sign of the real part of $\lambda_{\partial_x F(\cdot, \varepsilon)}(\psi^*(\varepsilon))$, given by the roots of the polynomial $p_\varepsilon(\lambda) = \det(\lambda I - \partial_x F(\psi^*(\varepsilon), \varepsilon))$. Since ψ^* is smooth, the coefficients of p_ε , and therefore its roots, depend continuously in ε . Hence, from (7), there is some $\varepsilon^* > 0$ such that $\text{Re}(\lambda_{\partial_x F(\cdot, \varepsilon)}(\psi^*(\varepsilon))) < 0$ for all $\varepsilon < \varepsilon^*$. [CM says: $\varepsilon^* \leq \underline{\varepsilon} \leq \varepsilon_{\text{IFT}}$ is implicit here. Not sure if it's worth it to spell it out. Also some abuse of notation (denoting the 4 eigenvalues by a single one), is it clear? Other than that, I think this one's done. It's just qualitative though, as ε^* might be large or small. It is possible to quantify ε_{IFT} in terms of the parameters, but we still don't have control on the other ε 's.] $\square$

4.1.1 Asymptotic approximation of I_ε^*

It is desirable to obtain an approximation of the endemic equilibrium for small values of ε . For this, we recall that the constraint for the incidence of the endemic equilibrium can be written as

$$P(I, \varepsilon) = P_0(I) + \varepsilon P_1(I) + \varepsilon^2 P_2(I) = 0, \quad (10)$$

where P_0 is a polynomial of degree 2 in I , while P_1 and P_2 are polynomials of degree less or equal than 4 in I . From [4] we know there is a unique $I_0^* \in]0, 1]$ such that

$$P_0(I_0^*) = a_0 + a_1 I_0^* + a_2 (I_0^*)^2 = 0. \quad (11)$$

Then

$$I_0^* = \frac{-a_1 \pm \sqrt{D}}{2a_2}, \quad D = a_1^2 - 4a_0a_2, \quad (12)$$

and

$$P_0'(I_0^*) = 2a_2 I_0^* + a_1 = \pm \sqrt{D}, \quad P_0''(I_0^*) = 2a_2, \quad P_0'''(I_0^*) = 0. \quad (13)$$

For a small enough ε , let's consider a (convergent Taylor) series representation of ψ from Proposition 4.2:

$$\psi(\varepsilon) = I_0^* + h(\varepsilon) := I_0^* + \sum_{k \in \mathbb{N}} h_k \varepsilon^k, \quad (14)$$

which we can plug in (10) to collect terms of increasing order in ε . Given any power series $f(z) = \sum_{k \in \mathbb{N}} f_k z^k$, we define the coefficient extractor operator $[z^n]$ [5, Ch. I.1] as

$$[z^n]f(z) = f_n. \quad (15)$$

[CM says: $[z^n]$ is linear and satisfies $[z^n](z^k f(z)) = [z^{n-k}]f(z)$]

Since P_0 , P_1 , and P_2 are polynomials, we can write

$$P_0(\psi) = P'_0(I_0^*)h + a_2h^2, \quad P_1(\psi) = \sum_{k=0}^4 \frac{h^k}{k!} P_1^{(k)}(I_0^*), \quad P_2(\psi) = \sum_{k=0}^4 \frac{h^k}{k!} P_2^{(k)}(I_0^*). \quad (16)$$

Substituting into (10) we get

$$P_0(\psi) = -\varepsilon P_1(\psi) - \varepsilon^2 P_2(\psi) \quad (17a)$$

$$\Rightarrow h = -P'_0(I_0)^{-1} \left(a_2h^2 + \varepsilon \sum_{k=0}^4 \frac{h^k}{k!} P_1^{(k)}(I_0^*) + \varepsilon^2 \sum_{k=0}^4 \frac{h^k}{k!} P_2^{(k)}(I_0^*) \right) \quad (17b)$$

$$\Rightarrow h_n = -P'_0(I_0)^{-1}[\varepsilon^n] \left(a_2h^2 + \varepsilon \sum_{k=0}^4 \frac{h^k}{k!} P_1^{(k)}(I_0^*) + \varepsilon^2 \sum_{k=0}^4 \frac{h^k}{k!} P_2^{(k)}(I_0^*) \right). \quad (17c)$$

Using (17), it is possible to compute some orders explicitly.

Order ε Note that every term in h^2 and $\varepsilon^2 P_2$ is already of order ≥ 2 . Similarly, for P_1 the only term that is of order less than 2 in the sum is the first one. Thus, we get

$$h_1 = -P'_0(I_0)^{-1} P_1(I_0^*). \quad (18)$$

Order ε^2 From a_2h^2 , the only term that survives is $a_2h_1\varepsilon^2$. From εP_1 , we only get $\varepsilon^2 h_1 P'_1(I_0^*)$ and for $\varepsilon^2 P_2$ we have $\varepsilon^2 P_2(I_0^*)$. Therefore

$$h_2 = a_2h_1 + h_1 P'_1(I_0^*) + P_2(I_0^*) = -P'_0(I_0)^{-1} P_1(I_0^*) (a_2 + P'_1(I_0^*)) + P_2(I_0^*). \quad (19)$$

[CM says: This is general and already a third order approximation, although we can keep going. The coefficients do not seem to simplify nicely. For the symbolic computations we don't really have to write the Taylor series for P_1 and P_2 in (17).]

4.2 Moderate mutation feedback (bifurcation region)

Figure 10 traces the endemic prevalence as a function of β_0 for several values of $\tilde{\varepsilon}$, for the transmission-driven variant. At small $\tilde{\varepsilon}$ (left panel), each curve rises from $\mathcal{R}_0 = 1$ and remains stable throughout. At larger $\tilde{\varepsilon}$ (right panel), each curve develops an unstable segment, bounded by two Hopf bifurcation points.

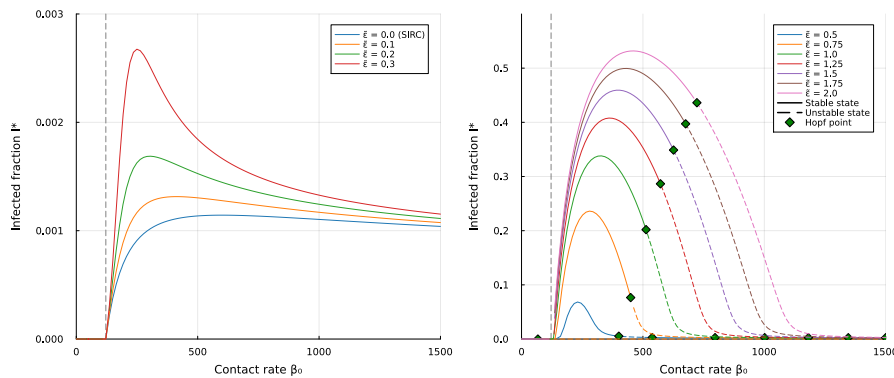


Figure 10: Endemic prevalence I^* of the transmission-driven variant against contact rate β_0 under varying mutation feedback strenghts ($\tilde{\varepsilon}$). Solid and dashed curves denote stable and unstable equilibria respectively. Note the scale difference in the y-axis.

[DR says: logarithmic axis and split axis didn't really work. i will try inset axis or leave it this way]

The analogous plot for the prevalence-drive variant (Figure 11) shows no such unstable segments, and each curve retains the same qualitative shape as the SIRC model, simply attaining a higher peak prevalence as $\tilde{\varepsilon}$ increases.

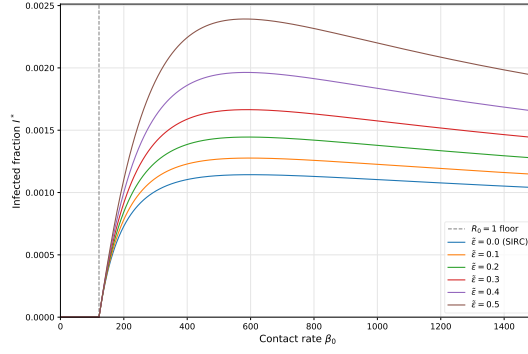


Figure 11: Endemic prevalence I^* against contact rate β_0 for different values of $\tilde{\epsilon}$

The eigenvalue sweep of Figure 5 was computed at a single, fixed pair of parameter values, $\beta_0 = 600$ and $\sigma = 0.07874$. Two natural questions follow: how do the two Hopf bifurcation points move as β_0 and σ vary, and does the qualitative three-regime picture persist across the ranges of these parameters relevant to influenza-like transmission? Figure 12 answers this by tracing the Hopf bifurcation curves via numerical continuation.

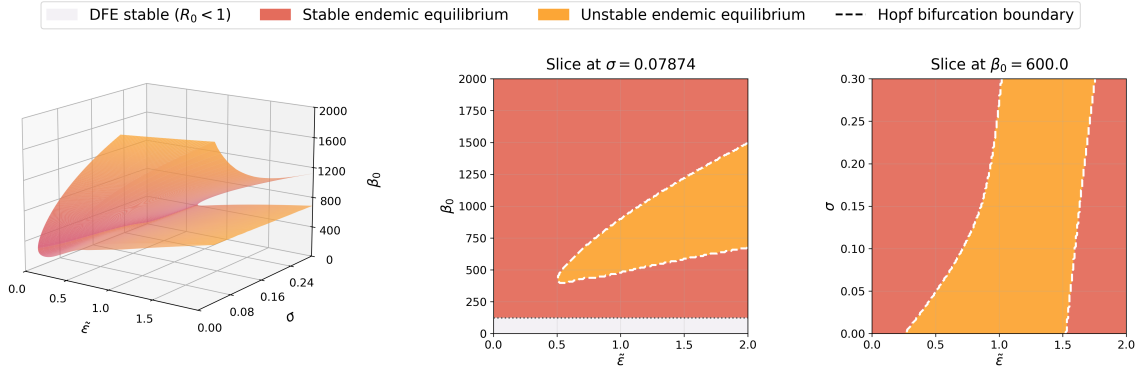


Figure 12: Stability regions of the endemic equilibrium of the transmission-driven variant across the three-parameter space $(\tilde{\epsilon}, \beta_0, \sigma)$, computed via numerical continuation of the Hopf bifurcation curve.

Figure 4 shows that the two Hopf bifurcation points along the diagonal at $\beta_0 = 600$ occur at $\tilde{\epsilon} \approx 0.63$ and $\tilde{\epsilon} \approx 1.58$. Figure 13 illustrates the resulting change in the dynamics of $I(t)$ directly. We select $\tilde{\epsilon} = 0.25$, well below the first bifurcation point, where the endemic equilibrium is stable and $I(t)$ settles into a fixed prevalence above SIRC after a few damped oscillations. We then take $\tilde{\epsilon} = 1.57$, just below the second bifurcation point, so the equilibrium is still unstable and the trajectory shows sustained, large-amplitude oscillations. Finally, we take $\tilde{\epsilon} = 1.59$ just past the second bifurcation point, where stability has only just been recovered and $I(t)$ decays again toward the endemic equilibrium.

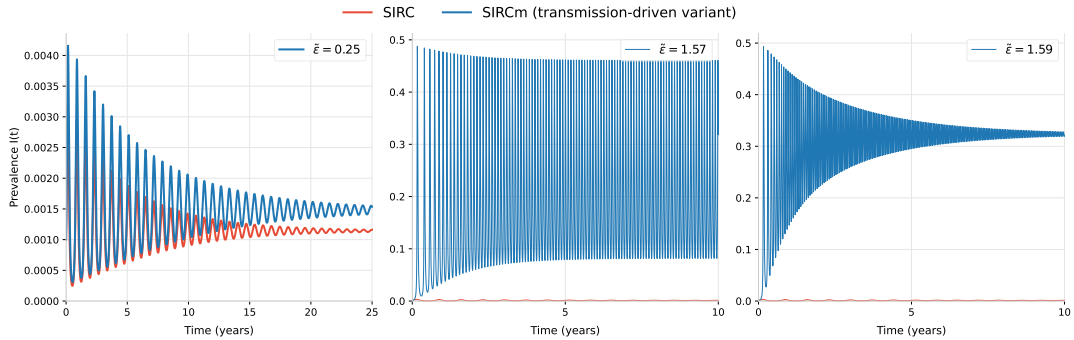


Figure 13: Time series of the infected fraction $I(t)$ for the transmission-driven variant at $\beta_0 = 600$, compared against SIRC.

For the prevalence-driven variant, no comparable bifurcation structure exists. Figure 3 shows that the diagonal $\tilde{\varepsilon}_1 = \tilde{\varepsilon}_2$ never enters the unstable region for any $\beta_0 \in [200, 2000]$. Figure 14 illustrates instead how the rate of convergence to equilibrium changes with feedback strength, using the same values of $\tilde{\varepsilon}$ as in Figure 8. As $\tilde{\varepsilon}$ increases, the endemic prevalence rises above its SIRC baseline, while convergence to the equilibrium becomes faster.

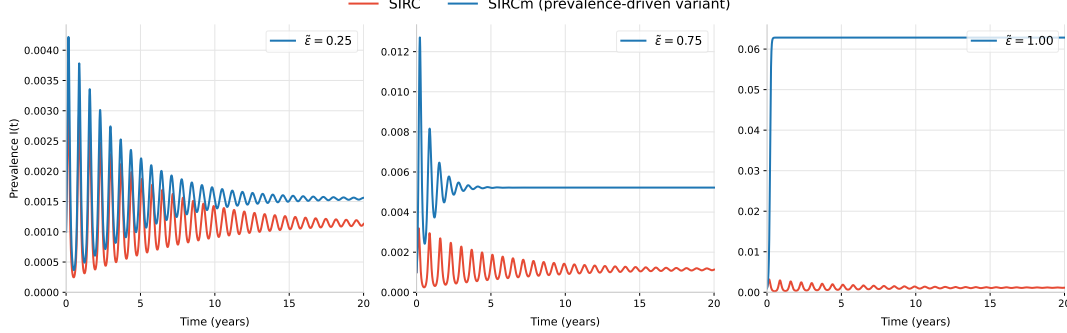


Figure 14: Time series of the infected fraction $I(t)$ for the prevalence-driven variant at $\beta_0 = 600$, compared against SIRC.

4.3 Strong mutation feedback (beyond the bifurcation region)

Let us study the behavior of the SIRCm model (1) as $\varepsilon \rightarrow +\infty$. We begin by recalling that for every ε the endemic equilibrium satisfies

$$R^* = \frac{\alpha I^* + (1 - \sigma)\beta C^* I^*}{\mu + (1 + \varepsilon g)\delta}, \quad C^* = \frac{\delta(1 + \varepsilon g)R^*}{\beta I^* + \mu + (1 + \varepsilon g)\gamma}. \quad (20)$$

Since the numerator of R^* is bounded, we have $R^* = \mathcal{O}(\varepsilon^{-1})$. [CM says: From here on I am just posing our problem in the framework of [6] to use Fenichel's Theorem as they state it. In this Section I need that g only depends on I and S] Moreover, for $\varepsilon \gg 1$ we have $C^* \approx \frac{\delta}{\gamma} R^* = \mathcal{O}(\varepsilon^{-1})$. This motivates the introduction of the scaled variables

$$R = \varepsilon^{-1} \bar{R}, \quad C = \varepsilon^{-1} \bar{C}, \quad (21)$$

which, from equations (1c) and (1d), gives us

$$\dot{S} = \mu(1 - S) - \beta SI + (\varepsilon^{-1} + g(x))\gamma \bar{C}, \quad (22a)$$

$$\dot{I} = \beta SI + \sigma \beta \varepsilon^{-1} \bar{C} I - (\mu + \alpha)I, \quad (22b)$$

$$\varepsilon^{-1} \dot{\bar{R}} = (1 - \sigma)\beta \varepsilon^{-1} \bar{C} I + \alpha I - \varepsilon^{-1} \mu \bar{R} - (\varepsilon^{-1} + g(x))\delta \bar{R}, \quad (22c)$$

$$\varepsilon^{-1} \dot{\bar{C}} = (\varepsilon^{-1} + g(x))\delta \bar{R} - \beta \varepsilon^{-1} \bar{C} I - \mu \varepsilon^{-1} \bar{C} - (\varepsilon^{-1} + g(x))\gamma \bar{C}, \quad (22d)$$

recalling that $g(x)$ only depends on S and I . Moreover, writing (22) as

$$(\dot{S}, \dot{I})^\top = f_{\text{slow}}(x, \varepsilon^{-1}), \quad \varepsilon^{-1} (\dot{\bar{R}}, \dot{\bar{C}})^\top = f_{\text{fast}}(x, \varepsilon^{-1}), \quad (23)$$

defining the fast time scale $\tau = t\varepsilon$, and denoting $y' = \frac{dy}{d\tau}$, we can express (22) as

$$(S', I')^\top = \varepsilon^{-1} f_{\text{slow}}(x, \varepsilon^{-1}), \quad (\bar{R}', \bar{C}')^\top = f_{\text{fast}}(x, \varepsilon^{-1}). \quad (24)$$

As $\varepsilon^{-1} \rightarrow 0$, the fast variables $\bar{R}$ and $\bar{C}$ become enslaved to the slow variables S and I , defining the critical manifold

$$\mathcal{M}_\infty = \left\{ x \in \mathbb{R}^4 : \bar{R} = \frac{\alpha I}{\delta g(x)}, \quad \bar{C} = \frac{\alpha I}{\gamma g(x)} \right\}, \quad (25)$$

in which the slow subsystem

$$\dot{S} = I[-\beta S + (\mu + \alpha)], \quad (26a)$$

$$\dot{I} = I[\beta S - (\mu + \alpha)], \quad (26b)$$

is defined (noting that $S + I = 1$ in $\mathcal{M}_\infty$). Note that (26) does not depend on g , and that points on $\mathcal{M}_\infty$ are equilibria of the fast subsystem

$$\bar{R}' = \alpha I - g(x)\delta\bar{R}, \quad (27)$$

$$\bar{C}' = g(x)(\delta\bar{R} - \gamma\bar{C}), \quad (28)$$

$$S' = I' = 0. \quad (29)$$

The endemic equilibrium of (26) is unique and given by

$$S_\infty^* = \frac{\mu + \alpha}{\beta} = \mathcal{R}_0^{-1}, \quad I_\infty^* = 1 - \mathcal{R}_0^{-1}. \quad (30)$$

The Jacobian of the right-hand side of (26) has eigenvalues

$$\lambda_\infty \in \{-\mu, -\alpha - \beta - \mu + 2\beta S\}. \quad (31)$$

Thus, as $\varepsilon \rightarrow +\infty$, the unique equilibrium of the slow subsystem will be stable if and only if $\mathcal{R}_0 > 1$.

Let us take $I_{\min} > 0, S_{\min} > 0$ small enough such that $I_\infty^* > I_{\min}$ and $S_\infty^* > S_{\min}$. [CM says: Tis is just to prevent g from vanishing...in the prevalence case it's clean because we can just take x_0^*]. Then, we define the compact, smooth, submanifold of $\mathcal{M}_\infty$,

$$\mathcal{K}_\infty = \{x \in \mathcal{M}_\infty : I \geq I_{\min}, S \geq S_{\min}\}. \quad (32)$$

Then, after computing the Jacobian of f_{fast} with respect to $\bar{R}$ and $\bar{C}$, we see that

$$(\partial_{\bar{R}, \bar{C}} f_{\text{fast}})(x, \varepsilon^{-1}) = \begin{pmatrix} -\varepsilon^{-1}\mu - (\varepsilon^{-1} + g(x))\delta & (1 - \sigma)\beta\varepsilon^{-1}I \\ (\varepsilon^{-1} + g(x))\delta & -\beta\varepsilon^{-1}I - \mu\varepsilon^{-1} - (\varepsilon^{-1} + g(x))\gamma \end{pmatrix}, \quad (33)$$

and thus, for $x \in \mathcal{K}_\infty$, the eigenvalues of $(\partial_{\bar{R}, \bar{C}} f_{\text{fast}})(x, 0)$ are given by

$$\lambda \in \{-g(x)\delta, -g(x)\gamma\}. \quad (34)$$

Since $g(x) > 0$ in $\mathcal{K}_\infty$, $\mathcal{K}_\infty$ is an attracting normally hyperbolic submanifold of $\mathcal{M}_\infty$. [CM says: Invoke Fenichel, which does not grant us the unique equilibria, but gives us information on the dynamics]

Remark 4.3 (Recovery of the classical SIS model). *Equations (26) are, in fact, the classical SIS model, where recovered individuals return directly to S with no intermediate immune class.*

4.3.1 A Fenichel-free approach

4.3.2 Asymptotic approximation of I_ε^*

It is possible to express I_ε^* as a series expansion of the form

$$I_\varepsilon^* \approx \phi(\varepsilon^{-1}) = I_\infty + q(\varepsilon^{-1}) = I_\infty + \sum_{k \in \mathbb{N}} q_k \varepsilon^{-k}. \quad (35)$$

It suffices to observe that, for every ε , the endemic equilibrium satisfies $\varepsilon^{-2}P_0(I) + \varepsilon^{-1}P_1(I) + P_2(I) = 0$, where $P_2(I)$ is a cubic polynomial in I whose only non-zero root is I_∞^* . Then, we can proceed analogously to Section 4.1.1 to compute the coefficients q_k . [CM says: ϕ here is the analytic function guaranteed to exist by the IFT, but it might not be necessary]

4.4 Seasonally forced dynamics

We replace the constant transmission rate β_0 with $\beta(t) = \beta_0(1 + \eta \cos(2\pi t))$, following the seasonal forcing scheme of Casagrandi et al. [4]. We restrict attention to the prevalence-driven variant with a single feedback parameter ($\varepsilon := \varepsilon_1 = \varepsilon_2$), since Section 4 established that this variant's unforced endemic

equilibrium remains stable for all $\tilde{\varepsilon}$ tested, mirroring the stability of the SIRC endemic equilibrium itself. This lets us isolate the effect of mutation feedback on the seasonally forced dynamics.

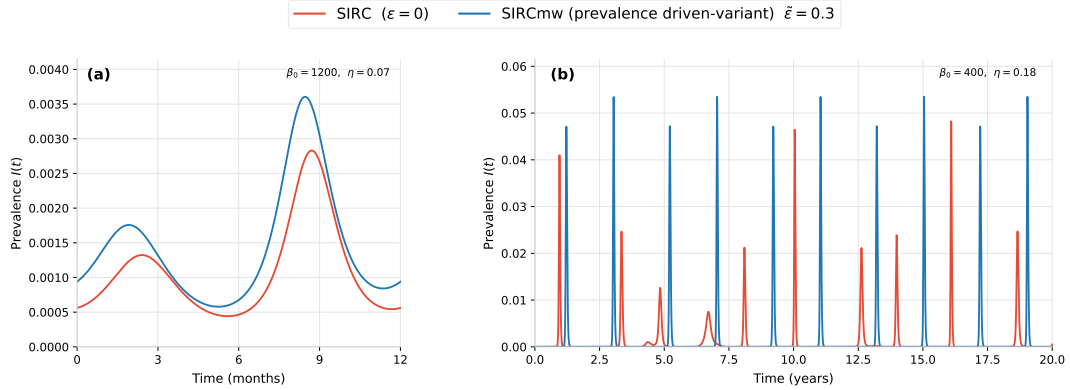


Figure 15: Replication and extension of Figure 5 in [4] for the prevalence-driven variant of the SIRCm model ($\tilde{\varepsilon} = 0.3$) compared to the standard SIRC baseline ($\tilde{\varepsilon} = 0$). The parameter η represents the degree of seasonality, renamed here to avoid confusion with the feedback strength.

[DR says: add data points? from the plots in casagrandi's paper]

Figure 15 replicates Figure 5 of Casagrandi et al. [4], comparing the SIRCm prevalence-driven variant ($\tilde{\varepsilon} = 0.3$) against the SIRC baseline ($\tilde{\varepsilon} = 0$) under the same two parameter regimes: panel (a) uses parameters representative of a tropical regime with low seasonality and high baseline transmission, while panel (b) uses those representative of a temperate regime with stronger seasonal forcing and lower baseline transmission.

In the tropical regime (a), both models settle into a stable annual cycle, with SIRCm attaining a higher peak prevalence than SIRC, consistent with the elevated endemic prevalence documented in Section 4. In the temperate regime (b), while SIRC settles into the chaotic attractor reported by Casagrandi et al. for this parameter setting, SIRCm converges to a period-2 cycle. These two regimes illustrate that mutation feedback can alter not only the level but also the qualitative character of the seasonally forced dynamics; further comparisons with different values of $\tilde{\varepsilon}$ are given in Appendix C.

5 Discussion and biological implications

The results of Section 4 show that mutation feedback changes the behaviour of the epidemic, producing sustained oscillation, a loss and recovery of stability, and differences in the two ways we modeled mutation supply.

It is useful to place SIRCm relative to two familiar reference points. When mutation feedback is switched off ($\varepsilon_1 = \varepsilon_2 = 0$), SIRCm is identical to the SIRC model of Casagrandi et al. [4], in which recovered hosts pass through a period of partial immunity (cross-immunity) before eventually becoming fully susceptible again. When mutation feedback is very strong, we show in Section 4 that this intermediate immune period effectively disappears, and hosts that recover are susceptible again almost immediately. In this case the model reduces to the classical SIS model of a disease that confers no immunity at all. SIRCm can thus be read as a model that interpolates between a pathogen against which the host retains meaningful immunity, and a pathogen against which the host's prior infection provides essentially no protection at all, because mutation degrades that protection as fast as it is acquired. [DR says: here look for applications of SIS model to discuss if the strong feedback is biologically realistic]

This interpolation is not gradual. Figures 2 and 16 show that the long-run fraction of the population that is infected at any given time, I^* , stays close to its SIRC value across a wide range of weak to moderate mutation feedback, and rises sharply once the feedback strength $\tilde{\varepsilon}$ passes a threshold, after which I^* climbs quickly toward its theoretical ceiling.

Another interesting finding of this work concerns what happens to the endemic state of the transmission-driven variant. Here, as mutation feedback strengthens through the moderate range, the steady and

constant level of infection loses stability, and begins to oscillate persistently, tracing out a repeating cycle over time. If feedback strengthens further still, stability returns and the system again settles to a constant level. This is notable because nothing in the model changes with time; there is no built-in seasonality and no external driver of any kind. The oscillation is generated entirely by the feedback loop between infection and mutation. This offers a different explanation for recurring epidemic waves than the usual one [8], which attributes them to external seasonal changes in transmission (e.g. people spending more time indoors in the winter).

The transmission-driven and prevalence-driven variants agree on the two-regime prevalence structure described above, but disagree on stability: the prevalence-driven variant shows no loss of stability along the diagonal $\varepsilon_1 = \varepsilon_2$ (Figure 3), while the transmission-driven variant does, once β_0 is large enough (Figure 4). The two variants correspond to different hypothesis about how mutation opportunities arise: at the point of transmission versus in proportion to the standing infected population regardless of transmission.

Acknowledgments

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A The equilibrium polynomial

Eliminating S, R, C from the equilibrium equations of SIRCm (1) yields the following polynomial in I

$$\mathcal{P}(I, \varepsilon_1, \varepsilon_2) = c_4 I^4 + c_3 I^3 + c_2 I^2 + c_1 I + c_0 = 0, \quad (36)$$

where

$$\begin{aligned} c_4 &= -\beta^2 \gamma \delta \mu \varepsilon_1 \varepsilon_2 (-\alpha^2 \gamma \varepsilon_2 + \alpha(\beta^2 \sigma + \gamma \delta \sigma \varepsilon_1 - \gamma \varepsilon_2 (\delta \sigma + 2\mu)) \\ &\quad + \beta \sigma (\beta (\delta \sigma + \mu) + \delta \mu \sigma \varepsilon_1) + \gamma \mu (\delta \sigma \varepsilon_1 - \varepsilon_2 (\delta \sigma + \mu))) \\ c_3 &= \beta \mu (\alpha^3 \gamma^2 \delta \varepsilon_1 \varepsilon_2^2 + \alpha^2 \gamma \varepsilon_2 (\beta^2 (\gamma \varepsilon_2 + \delta (\varepsilon_1 - 2\sigma \varepsilon_1)) - \beta \gamma \delta \varepsilon_1 \varepsilon_2 + \gamma \delta \varepsilon_1 (4\mu \varepsilon_2 - \delta (\sigma + 1) (\varepsilon_1 - \varepsilon_2))) \\ &\quad + \alpha (\beta^4 (\sigma - 1) (\delta \sigma \varepsilon_1 - \gamma \varepsilon_2) + \beta^3 \gamma \delta \sigma \varepsilon_1 \varepsilon_2 + \beta^2 \gamma (\gamma \varepsilon_2 (\delta (-2\sigma \varepsilon_1 + \varepsilon_1 + \sigma \varepsilon_2 + \varepsilon_2) + 2\mu \varepsilon_2) \\ &\quad + \delta \varepsilon_1 (\delta (\sigma - 1) \sigma \varepsilon_1 - 2\varepsilon_2 (\delta \sigma^2 + 3\mu \sigma - \mu))) + \beta \gamma \delta \varepsilon_1 \varepsilon_2 (2\gamma \delta \sigma \varepsilon_1 - \gamma \varepsilon_2 (2\delta \sigma + \mu (\sigma + 2)) \\ &\quad - \delta \mu (\sigma + 1) \sigma \varepsilon_1) + \gamma^2 \delta \mu \varepsilon_1 \varepsilon_2 (5\mu \varepsilon_2 - 2\delta (\sigma + 1) (\varepsilon_1 - \varepsilon_2))) + \beta^4 (\sigma - 1) (\delta \sigma + \mu) (\delta \sigma \varepsilon_1 - \gamma \varepsilon_2) \\ &\quad + \beta^3 \delta \sigma \varepsilon_1 (\gamma \varepsilon_2 (2\delta \sigma - \mu (\sigma - 3)) + \delta \mu (\sigma - 1) \sigma \varepsilon_1) + \beta^2 \gamma (\gamma \varepsilon_2 (\delta^2 \sigma (\varepsilon_2 - \varepsilon_1) \\ &\quad + \delta \mu (-2\sigma \varepsilon_1 + \varepsilon_1 + \sigma \varepsilon_2 + \varepsilon_2) + \mu^2 \varepsilon_2) + \delta \mu \varepsilon_1 (\delta (\sigma - 1) \sigma \varepsilon_1 + \varepsilon_2 (\delta \sigma^2 (2\varepsilon_1 - 3) - 4\mu \sigma + \mu))) \\ &\quad + \beta \gamma \delta \mu \varepsilon_1 \varepsilon_2 (2\gamma \delta \sigma \varepsilon_1 - \gamma \varepsilon_2 (2\delta \sigma + \mu \sigma + \mu) - \delta \mu (\sigma + 1) \sigma \varepsilon_1) + \gamma^2 \delta \mu^2 \varepsilon_1 \varepsilon_2 (2\mu \varepsilon_2 - \delta (\sigma + 1) (\varepsilon_1 - \varepsilon_2))) \\ c_2 &= \mu (\alpha^3 \gamma^2 \delta \varepsilon_1 \varepsilon_2 (\delta (\varepsilon_2 - \varepsilon_1) + \mu \varepsilon_2) - \alpha^2 \gamma (\beta^2 (\delta \varepsilon_2 (\gamma \sigma \varepsilon_1 - 2\gamma \varepsilon_2 + 4\mu \sigma \varepsilon_1 - 2\mu \varepsilon_1) - 2\gamma \mu \varepsilon_2^2 \\ &\quad + \delta^2 \varepsilon_1 (-\sigma \varepsilon_1 + \varepsilon_1 + 2\sigma \varepsilon_2 - \varepsilon_2)) + \beta \delta \varepsilon_1 \varepsilon_2 (\delta \mu \sigma \varepsilon_1 - \gamma (\sigma + 1) (\delta \varepsilon_1 - \delta \varepsilon_2 - \mu \varepsilon_2)) \\ &\quad + 3\gamma \delta \mu \varepsilon_1 \varepsilon_2 (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2)) + \alpha (\beta^4 (\sigma - 1) (-\gamma^2 \varepsilon_2 + \gamma \delta (\sigma \varepsilon_1 + \varepsilon_1 - 2\varepsilon_2) - 3\gamma \mu \varepsilon_2 \\ &\quad + \delta \sigma \varepsilon_1 (\delta + 3\mu)) + \beta^3 (\gamma^2 (-\varepsilon_2) (\delta (-2\sigma \varepsilon_1 + \varepsilon_1 + \sigma \varepsilon_2 + \varepsilon_2) + \mu (\sigma + 1) \varepsilon_2) + \gamma \delta \varepsilon_1 (-\delta (\sigma - 1) \sigma \varepsilon_1 \\ &\quad + 2\delta \sigma^2 \varepsilon_2 + \mu (\sigma^2 + 3\sigma - 1) \varepsilon_2) + \delta^2 \mu (\sigma - 1) \sigma \varepsilon_1^2) + \beta^2 \gamma (\gamma (-\delta^2 (\varepsilon_1 - \varepsilon_2) (\sigma \varepsilon_1 (\varepsilon_2 - 1) + \varepsilon_1 \\ &\quad + (\sigma + 1) \varepsilon_2)) + \delta \mu \varepsilon_2 (\sigma \varepsilon_1 (\varepsilon_2 - 4) + \varepsilon_1 + (\sigma + 5) \varepsilon_2) + 4\mu^2 \varepsilon_2^2) + \delta \mu \varepsilon_1 (\delta \varepsilon_1 (\sigma^2 (\varepsilon_2 + 1) + \sigma (\varepsilon_2 + 1) \\ &\quad - 2) - 2\delta (\sigma^2 + 2\sigma - 1) \varepsilon_2 + \mu (4 - 9\sigma) \varepsilon_2)) + 2\beta \gamma \delta \mu \varepsilon_1 \varepsilon_2 (\gamma (\sigma + 1) (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) - \delta \mu \sigma \varepsilon_1) \\ &\quad + 3\gamma^2 \delta \mu^2 \varepsilon_1 \varepsilon_2 (\delta (\varepsilon_2 - \varepsilon_1) + \mu \varepsilon_2)) + \beta^5 (\sigma - 1) (\delta \sigma + \mu) (\gamma \varepsilon_2 - \delta \sigma \varepsilon_1) - \beta^4 (\gamma^2 (\sigma - 1) \varepsilon_2 (\delta + \mu) \\ &\quad + \gamma (\delta^2 \sigma (\varepsilon_1 (\sigma (\varepsilon_2 - 2) + 2) + (\sigma - 1) \varepsilon_2) + \delta \mu (2(\sigma^2 - 1) \varepsilon_2 - \varepsilon_1 (\sigma^2 + (\sigma - 2) \sigma \varepsilon_2 - 1))) \\ &\quad + 3\mu^2 (\sigma - 1) \varepsilon_2) + \delta \mu (\sigma - 1) \sigma \varepsilon_1 (\delta (\sigma (\varepsilon_1 - 2) - 1) - 3\mu)) + \beta^3 (\gamma^2 (-\varepsilon_2) (2\delta^2 \sigma (\varepsilon_2 - \varepsilon_1) \\ &\quad + \delta \mu (-3\sigma \varepsilon_1 + \varepsilon_1 + 3\sigma \varepsilon_2 + \varepsilon_2) + \mu^2 (\sigma + 1) \varepsilon_2) - \gamma \delta \mu \varepsilon_1 \varepsilon_2 (\delta \sigma^2 (\varepsilon_1 - 4) - 5\mu \sigma + \mu) \\ &\quad + \delta^2 \mu^2 \sigma (\sigma^2 - 1) \varepsilon_1^2) + \beta^2 \gamma \mu (\gamma (-\delta^2 (\varepsilon_1 - \varepsilon_2) (\sigma \varepsilon_1 (\varepsilon_2 - 1) + \varepsilon_1 + (\sigma + 1) \varepsilon_2)) + \delta \mu \varepsilon_2 (\sigma \varepsilon_1 (\varepsilon_2 - 3) \\ &\quad + \varepsilon_1 + (\sigma + 3) \varepsilon_2) + 2\mu^2 \varepsilon_2^2) + \delta \mu \varepsilon_1 (\delta ((\sigma + 1) \varepsilon_1 (\sigma + \sigma \varepsilon_2 - 1) - 2\sigma^2 \varepsilon_2 - 2\sigma \varepsilon_2 + \varepsilon_2) \\ &\quad + \mu (2 - 5\sigma) \varepsilon_2)) + \beta \gamma \delta \mu^2 \varepsilon_1 \varepsilon_2 (\gamma (\sigma + 1) (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) - \delta \mu \sigma \varepsilon_1) + \gamma^2 \delta \mu^3 \varepsilon_1 \varepsilon_2 (\delta (\varepsilon_2 - \varepsilon_1) + \mu \varepsilon_2)) \\ c_1 &= \beta \mu (\alpha^2 \gamma (\gamma (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) (\mu (-\varepsilon_2) - \delta (-\sigma \varepsilon_1 + \varepsilon_1 + \varepsilon_2)) + \delta \mu \varepsilon_1 (\delta ((\sigma - 1) \varepsilon_1 - 2\sigma \varepsilon_2 + \varepsilon_2) \\ &\quad + \mu (1 - 2\sigma) \varepsilon_2)) + \alpha (\beta^2 (\sigma - 1) (\gamma^2 (\delta (\varepsilon_1 - 2\varepsilon_2) - 2\mu \varepsilon_2) + \gamma (\delta^2 (\sigma \varepsilon_1 + \varepsilon_1 - \varepsilon_2) \\ &\quad + 2\delta \mu (\sigma \varepsilon_1 + \varepsilon_1 - 2\varepsilon_2) - 3\mu^2 \varepsilon_2) + \delta \mu \sigma \varepsilon_1 (2\delta + 3\mu)) + \beta (-\gamma^2 (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) (\delta (\sigma - 1) \varepsilon_1 \\ &\quad - \delta (\sigma + 1) \varepsilon_2 - \mu (\sigma + 1) \varepsilon_2)) + \gamma \delta \mu \varepsilon_1 (\delta (\sigma - 1)^2 \varepsilon_1 + \delta (\sigma^2 + 2\sigma - 1) \varepsilon_2 + \mu (\sigma^2 + 2\sigma - 1) \varepsilon_2) \\ &\quad + \delta^2 \mu^2 (\sigma - 1) \sigma \varepsilon_1^2) + 2\gamma \mu (\gamma (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) (\mu (-\varepsilon_2) - \delta (-\sigma \varepsilon_1 + \varepsilon_1 + \varepsilon_2)) + \delta \mu \varepsilon_1 (\delta ((\sigma - 1) \varepsilon_1 \\ &\quad - 2\sigma \varepsilon_2 + \varepsilon_2) + \mu (1 - 2\sigma) \varepsilon_2))) + \beta^3 (\sigma - 1) (\gamma^2 \varepsilon_2 (\delta + \mu) + \gamma (\delta^2 \sigma (\varepsilon_2 - 2\varepsilon_1) + \delta \mu ((\sigma + 2) \varepsilon_2 \\ &\quad - (\sigma + 1) \varepsilon_1) + 2\mu^2 \varepsilon_2) - \delta \mu \sigma \varepsilon_1 (\delta \sigma + \delta + 2\mu)) - \beta^2 (\gamma^2 (\delta + \mu) (\delta (\varepsilon_1 - \varepsilon_2) (\sigma (\varepsilon_2 - 1) + 1) \\ &\quad - \mu \varepsilon_2 (\sigma (\varepsilon_2 - 2) + 2)) + \gamma \mu (\delta^2 ((\sigma - 1) \sigma \varepsilon_1^2 + 2\sigma \varepsilon_1 + \sigma^2 \varepsilon_1 (\varepsilon_2 - 3) + \varepsilon_1 + (\sigma^2 - 1) \varepsilon_2) \\ &\quad + \delta \mu (\varepsilon_1 (\sigma^2 (\varepsilon_2 - 2) + 2) + (\sigma^2 + 3\sigma - 4) \varepsilon_2) + 3\mu^2 (\sigma - 1) \varepsilon_2) + \delta \mu^2 (\sigma - 1) \sigma \varepsilon_1 (\delta (-\sigma + \varepsilon_1 - 2) \\ &\quad - 3\mu)) + \beta \mu (-\gamma^2 (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) (\delta (\sigma - 1) \varepsilon_1 - \delta (\sigma + 1) \varepsilon_2 - \mu (\sigma + 1) \varepsilon_2)) + \gamma \delta \mu \varepsilon_1 (\delta (\sigma - 1)^2 \varepsilon_1 \\ &\quad + \delta (\sigma^2 + 2\sigma - 1) \varepsilon_2 + \mu (\sigma^2 + 2\sigma - 1) \varepsilon_2) + \delta^2 \mu^2 (\sigma - 1) \sigma \varepsilon_1^2) + \gamma \mu^2 (\gamma (\delta (\varepsilon_1 - \varepsilon_2) - \mu \varepsilon_2) (\mu (-\varepsilon_2) \\ &\quad - \delta (-\sigma \varepsilon_1 + \varepsilon_1 + \varepsilon_2)) + \delta \mu \varepsilon_1 (\delta ((\sigma - 1) \varepsilon_1 - 2\sigma \varepsilon_2 + \varepsilon_2) + \mu (1 - 2\sigma) \varepsilon_2))) \\ c_0 &= \beta^2 \mu (\sigma - 1) (\gamma + \mu) (\delta + \mu) (\alpha - \beta + \mu) (\gamma \delta (\varepsilon_1 - \varepsilon_2) - \gamma \mu \varepsilon_2 + \delta \mu \sigma \varepsilon_1). \end{aligned}$$

when setting $\varepsilon_1 = \varepsilon_2 = \varepsilon$, we get a simpler 4 degree polynomial

$$\mathcal{P}(I, \varepsilon) = c_4 I^4 + c_3 I^3 + c_2 I^2 + c_1 I + c_0 = 0, \quad (37)$$

where

$$\begin{aligned} c_4 &= \beta^2 \gamma \delta \mu \varepsilon (-\alpha^2 \gamma \varepsilon + \alpha(\beta^2 \sigma - 2\gamma \mu \varepsilon) + \beta \sigma(\beta \delta \sigma + \beta \mu + \delta \mu \sigma \varepsilon) - \gamma \mu^2 \varepsilon) \\ c_3 &= \beta \mu (-\alpha^3 \gamma^2 \delta \varepsilon^2 - \alpha^2 \gamma \varepsilon (\beta^2 (\gamma - 2\delta \sigma + \delta) - \beta \gamma \delta \varepsilon + 4\gamma \delta \mu \varepsilon) + \alpha(\beta^4 (\sigma - 1)(\gamma - \delta \sigma) - \beta^3 \gamma \delta \sigma \varepsilon \\ &\quad + \beta^2 \gamma \varepsilon (\gamma \delta (\sigma - 2) - 2\gamma \mu + \delta^2 \sigma (\sigma + 1) + \delta \mu (6\sigma - 2)) + \beta \gamma \delta \mu \varepsilon^2 (\gamma (\sigma + 2) + \delta \sigma (\sigma + 1)) \\ &\quad - 5\gamma^2 \delta \mu^2 \varepsilon^2) + \beta^4 (\sigma - 1)(\gamma - \delta \sigma)(\delta \sigma + \mu) + \beta^3 \delta \sigma \varepsilon (-2\gamma \delta \sigma + \gamma \mu (\sigma - 3) - \delta \mu (\sigma - 1) \sigma) \\ &\quad + \beta^2 \gamma \mu \varepsilon (\gamma \delta (\sigma - 2) - \gamma \mu + \delta^2 \sigma (1 - 2\sigma (\varepsilon - 1)) + \delta \mu (4\sigma - 1)) + \beta \gamma \delta \mu^2 (\sigma + 1) \varepsilon^2 (\gamma + \delta \sigma) \\ &\quad - 2\gamma^2 \delta \mu^3 \varepsilon^2) \\ c_2 &= \mu (-\alpha^3 \gamma^2 \delta \mu \varepsilon^2 + \alpha^2 \gamma \varepsilon (\beta^2 (\gamma \delta (\sigma - 2) - 2\gamma \mu + \delta^2 \sigma + \delta \mu (4\sigma - 2)) + \beta \delta \mu \varepsilon (\gamma \sigma + \gamma + \delta \sigma) - 3\gamma \delta \mu^2 \varepsilon) \\ &\quad + \alpha(\beta^4 (\sigma - 1)(\gamma + \delta + 3\mu)(\gamma - \delta \sigma) - \beta^3 \varepsilon (\gamma^2 (\delta (\sigma - 2) - \mu (\sigma + 1)) + \gamma \delta (\delta \sigma (\sigma + 1) \\ &\quad + \mu (\sigma^2 + 3\sigma - 1)) + \delta^2 \mu (\sigma - 1) \sigma) - \beta^2 \gamma \mu \varepsilon (\gamma \delta (\sigma (\varepsilon - 3) + 6) + 4\gamma \mu + \delta^2 \sigma (-\sigma + \sigma \varepsilon + \varepsilon - 3) \\ &\quad + \delta \mu (4 - 9\sigma)) + 2\beta \gamma \delta \mu^2 \varepsilon^2 (\gamma \sigma + \gamma + \delta \sigma) - 3\gamma^2 \delta \mu^3 \varepsilon^2) + \beta^5 (\sigma - 1)(\delta \sigma - \gamma)(\delta \sigma + \mu) \\ &\quad + \beta^4 (\gamma^2 (\sigma - 1)(\delta + \mu) + \gamma (\delta^2 \sigma (\sigma (\varepsilon - 1) + 1) - \delta \mu (-\sigma^2 + (\sigma - 2) \sigma \varepsilon + 1) + 3\mu^2 (\sigma - 1)) \\ &\quad + \delta \mu (\sigma - 1) \sigma (\delta (\sigma (\varepsilon - 2) - 1) - 3\mu)) + \beta^3 \mu \varepsilon (\gamma^2 (2\delta + \mu \sigma + \mu) + \gamma \delta (\delta \sigma^2 (\varepsilon - 4) - 5\mu \sigma + \mu) \\ &\quad - \delta^2 \mu \sigma (\sigma^2 - 1)) - \beta^2 \gamma \mu^2 \varepsilon (\gamma \delta (\sigma (\varepsilon - 2) + 4) + 2\gamma \mu + \delta^2 \sigma (-\sigma + \sigma \varepsilon + \varepsilon - 2) + \delta \mu (2 - 5\sigma)) \\ &\quad + \beta \gamma \delta \mu^3 \varepsilon^2 (\gamma \sigma + \gamma + \delta \sigma) - \gamma^2 \delta \mu^4 \varepsilon^2) \\ c_1 &= -\beta \mu (\alpha^2 \gamma \mu \varepsilon (\gamma (\mu - \delta (\sigma - 2)) + \delta (-\delta \sigma - 2\mu \sigma + \mu)) - \alpha(\beta^2 (\sigma - 1)(\gamma (\delta + 2\mu) + \mu (2\delta + 3\mu))(\gamma - \delta \sigma) \\ &\quad + \beta \mu \varepsilon (\gamma^2 (2\delta + \mu \sigma + \mu) - \gamma \delta (2\delta \sigma^2 + \mu (\sigma^2 + 2\sigma - 1)) - \delta^2 \mu (\sigma - 1) \sigma) + 2\gamma \mu^2 \varepsilon (\gamma \delta (\sigma - 2) \\ &\quad - \gamma \mu + \delta^2 \sigma + \delta \mu (2\sigma - 1))) + \beta^3 (\sigma - 1)(\gamma - \delta \sigma)(\gamma (\delta + \mu) + \mu (\delta \sigma + \delta + 2\mu)) \\ &\quad + \beta^2 \mu (\gamma^2 (\delta + \mu)(\sigma (\varepsilon - 2) + 2) + \gamma (\delta^2 \sigma (2\sigma - 2\sigma \varepsilon + \varepsilon - 2) + \delta \mu (-3\sigma - (\sigma^2 (\varepsilon - 1)) + 2) \\ &\quad - 3\mu^2 (\sigma - 1)) + \delta \mu (\sigma - 1) \sigma (\delta (\sigma - \varepsilon + 2) + 3\mu)) - \beta \mu^2 \varepsilon (\gamma^2 (2\delta + \mu \sigma + \mu) - \gamma \delta (2\delta \sigma^2 \\ &\quad + \mu (\sigma^2 + 2\sigma - 1)) - \delta^2 \mu (\sigma - 1) \sigma) + \gamma \mu^3 \varepsilon (\gamma (\mu - \delta (\sigma - 2)) + \delta (-\delta \sigma - 2\mu \sigma + \mu))) \\ c_0 &= \beta^2 \mu^2 (\sigma - 1)(\gamma + \mu)(\delta + \mu)(\alpha - \beta + \mu)(\gamma - \delta \sigma) \end{aligned}$$

B Prevalence-driven variant

B.0.1 Larger (or at least quantifiable) uniqueness region for the endemic equilibrium

Given $\varepsilon > 0$, the algebraic equation for the incidence of the endemic equilibrium reduces to

$$P(I) = a_3 I^3 + a_2 I^2 + a_1 I + a_0 = 0, \quad (38)$$

where

$$\begin{aligned} a_3 &= \beta \delta \varepsilon (\gamma \varepsilon + \beta \sigma), \\ a_2 &= \alpha(\beta + \gamma \varepsilon)(\beta + \delta \varepsilon) + \gamma \delta \varepsilon^2 \mu + \beta^2 (\mu - \delta (\varepsilon - 1) \sigma) + \beta \varepsilon (\gamma (2\delta - \delta \varepsilon + \mu) + \delta \mu (1 + \sigma)), \\ a_1 &= \delta \varepsilon \mu^2 + \gamma \varepsilon \mu (2\delta + \mu) + \alpha \beta (\gamma + \delta + 2\mu) + \beta \gamma (\delta - 2\delta \varepsilon + \mu - \varepsilon \mu) \\ &\quad + \alpha \varepsilon (\delta \mu + \gamma (2\delta + \mu)) - \beta^2 (\mu + \delta \sigma) + \beta \mu (2\mu + \delta (1 - \varepsilon + \sigma)) \\ &\quad + \beta \gamma (\delta (1 - 2\varepsilon) + \mu (1 - \varepsilon)) + \beta \mu (2\mu + \delta (1 - \varepsilon + \sigma)) - \beta^2 (\mu + \delta \sigma), \\ a_0 &= (\alpha - \beta + \mu)(\gamma + \mu)(\delta + \mu). \end{aligned}$$

Descartes' rule of signs states that the number of sign changes in the coefficients of P will bound the number of positive roots from above. Since $\mathcal{R}_0 = \frac{\beta}{\alpha + \mu} > 1$, $a_0 < 0$ and, as long as $\varepsilon > 0$, we also have $a_3 > 0$. Finally, it suffices to require $\varepsilon \leq 1$ for $a_2 > 0$. Then, P will have a unique positive root I^* (note

that the sign of a_1 is irrelevant as there will be a single sign change in any case). From the existence results in Section 3.1, we know that $x^* \in \Omega$. In particular, under the scaling (3), the additional condition for uniqueness becomes

$$\tilde{\varepsilon} \leq I(0), \quad (39)$$

where $I(0)$ is the incidence of the endemic equilibrium of SIRC.

B.0.2 Bifurcation and stability analysis

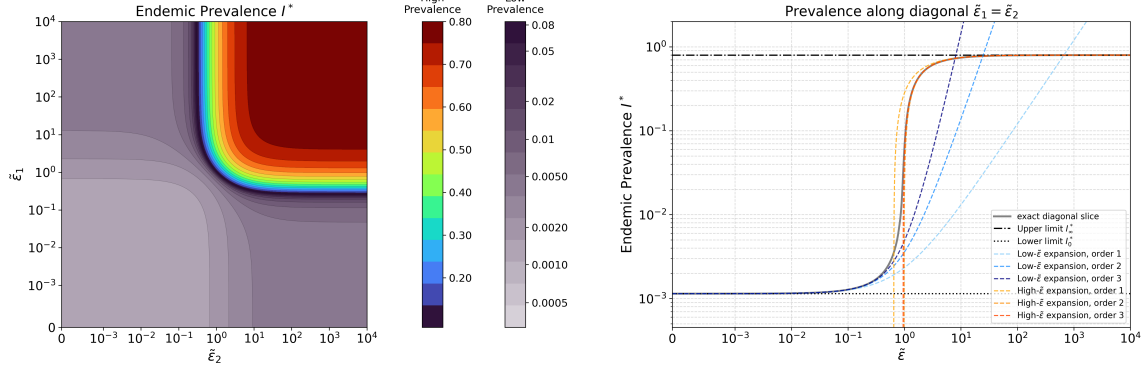


Figure 16: Endemic prevalence I^* of the prevalence-driven variant of the SIRCm model, analogous to Figure 2. *Left*: two-dimensional plot of I^* as a function of ε_1 and ε_2 . *Right*: diagonal slice $\tilde{\varepsilon}_1 = \tilde{\varepsilon}_2$, together with the first three orders of the Taylor series approximations for the low- and high-feedback regimes. The dotted and dash-dotted lines mark the lower and upper limits I_0^* and I_∞^* , respectively, as defined in the caption of Figure 2.

C More seasonally forced time series examples

These figures provide additional examples of the seasonally forced comparison introduced in Section 4.4. We contrast the tropical regime ($\beta_0 = 1200$, $\eta = 0.07$) against the temperate regime ($\beta_0 = 400$, $\eta = 0.18$), using the same SIRC baseline ($\tilde{\varepsilon} = 0$) for reference.

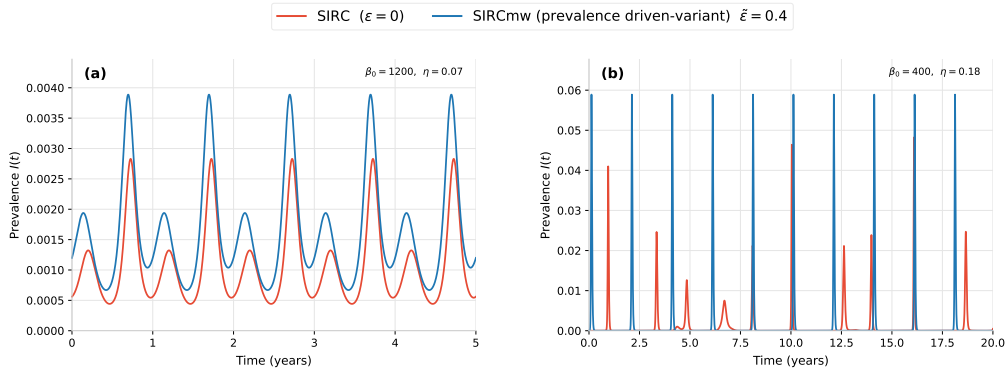


Figure 17: Replication and extension of Figure 5 in [4] for the prevalence-driven variant of the SIRCm model ($\tilde{\varepsilon} = 0.4$) compared to the standard SIRC baseline ($\tilde{\varepsilon} = 0$). In the tropical regime (a), both models remain on an annual cycle, with SIRCm attaining higher peak and trough prevalence than SIRC. In the temperate regime (b), SIRC settles into the chaotic attractor reported by Casagrandi et al., while SIRCm shows a regular periodic cycle.

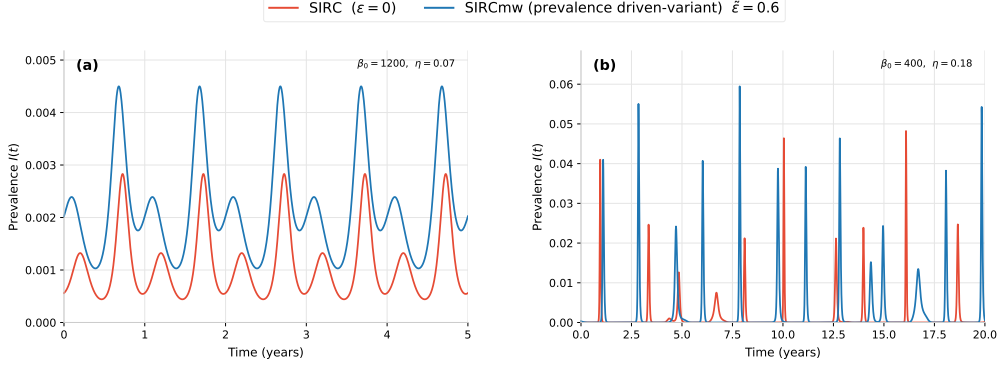


Figure 18: As Figure 17, at $\tilde{\varepsilon} = 0.6$. In the tropical regime (a), both models remain on an annual cycle, with SIRCm attaining higher peak and trough prevalence than SIRC. In the temperate regime (b), both models show chaotic behavior.

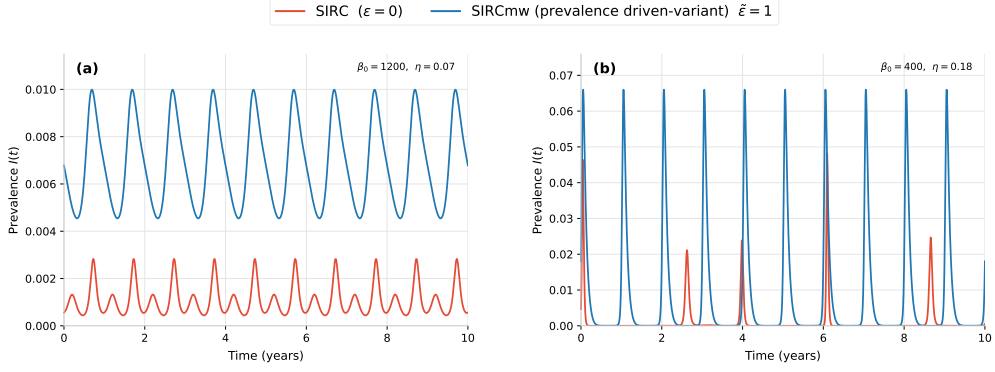


Figure 19: As Figure 17, at $\tilde{\varepsilon} = 1$. In the tropical regime (a), SIRCm now oscillates as a single-peaked annual cycle at substantially higher prevalence. In the temperate regime (b), SIRCm settles into a regular annual cycle, in contrast with the chaotic SIRC baseline.

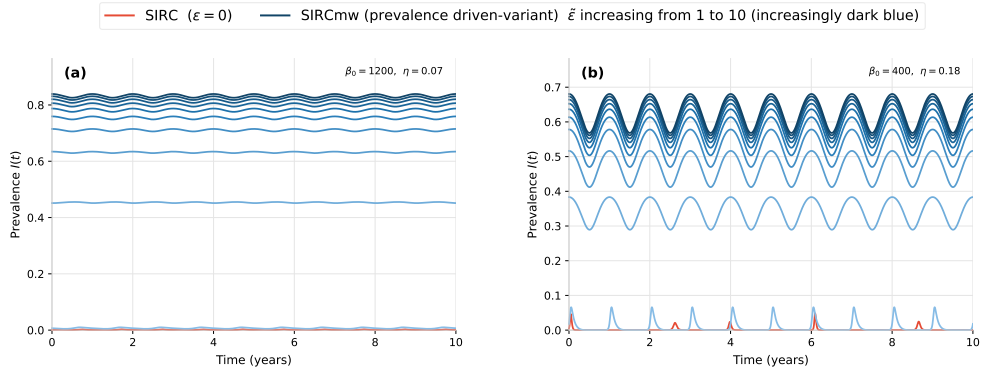


Figure 20: Family of prevalence-driven SIRCm time series for $\tilde{\varepsilon}$ ranging from 1 to 10 (increasingly dark blue), compared against the SIRC baseline ($\tilde{\varepsilon} = 0$, negligible at this scale). In both regimes, SIRCm remains on a smooth, single-peaked annual cycle throughout, and its mean prevalence rises toward the theoretical ceiling $I_{\infty}^* = 1 - \mathcal{R}_0^{-1}$ as $\tilde{\varepsilon}$ increases. In the tropical regime (a) the oscillation amplitude collapses, leaving an almost constant high prevalence. In the temperate regime (b) a visible annual oscillation persists, with the curves instead compressing toward the upper end of the range as $\tilde{\varepsilon}$ grows.

D Old approach asymptotic expansion large ε

As done in Section ??, we are interested in obtaining an approximation of the endemic equilibrium as $\varepsilon \rightarrow \infty$. We begin by recalling that for every ε the endemic equilibrium satisfies

$$R^* = \frac{\alpha I^* + (1 - \sigma)\beta C^* I^*}{\mu + (1 + \varepsilon g)\delta}, \quad C^* = \frac{\delta(1 + \varepsilon g)R^*}{\beta I^* + \mu + (1 + \varepsilon g)\gamma}. \quad (40)$$

Since the numerator of R^* is bounded, we have $R^* = \mathcal{O}(\varepsilon^{-1})$. Moreover, for $\varepsilon \gg 1$ we have $C^* \approx \frac{\delta}{\gamma} R^* = \mathcal{O}(\varepsilon^{-1})$. Therefore, noting that

$$\mu + (1 + \varepsilon g)\delta = (1 + \varepsilon g)\delta \left(1 + \frac{\mu}{(1 + g\varepsilon)\delta}\right) = (1 + \varepsilon g)\delta(1 + \mathcal{O}(\varepsilon^{-1})), \quad (41)$$

we obtain

$$R^* = \frac{\alpha I^*(1 + \mathcal{O}(\varepsilon^{-1}))}{(1 + \varepsilon g)\delta(1 + \mathcal{O}(\varepsilon^{-1}))} = \frac{\alpha I^*}{(1 + \varepsilon g)\delta}(1 + \mathcal{O}(\varepsilon^{-1})) = \frac{\alpha I^*}{\delta\varepsilon g} + \mathcal{O}(\varepsilon^{-2}). \quad (42)$$

Similarly, for C^* we begin by noting that

$$\frac{\delta(1 + \varepsilon)}{\beta I^* + \mu + (1 + \varepsilon g)\gamma} - \frac{\delta}{\gamma} = \frac{-\delta(\beta I^* + \mu)}{\beta\gamma I^* + \mu\gamma + (1 + \varepsilon g)\gamma^2} = \mathcal{O}(\varepsilon^{-1}), \quad (43)$$

and thus $C^* = \frac{\delta}{\gamma} R^* + \mathcal{O}(\varepsilon^{-2})$. [CM says: For most of these estimates we need $g(x^*)$ bounded away from zero uniformly.]

Finally, Equation (5) yields

$$I^* = (1 - \mathcal{R}_0^{-1}) - \frac{\alpha I^*}{\delta\varepsilon g} - (1 - \sigma)\frac{\alpha I^*}{\gamma\varepsilon g} + \mathcal{O}(\varepsilon^{-2}). \quad (44)$$

In the *prevalence-driven* case, we immediately get

$$I^* = (1 - \mathcal{R}_0^{-1}) - \frac{\alpha}{\delta\varepsilon} - (1 - \sigma)\frac{\alpha}{\gamma\varepsilon} + \mathcal{O}(\varepsilon^{-2}). \quad (45)$$

For the *transmission-driven* one, we use the relation (4) to get

$$S^* = \mathcal{R}_0^{-1} - \frac{\sigma\delta}{\gamma} R^* + \mathcal{O}(\varepsilon^{-2}) = \mathcal{R}_0^{-1} - \frac{\sigma\alpha}{\gamma\varepsilon S^*} + \mathcal{O}(\varepsilon^{-2}), \quad (46)$$

which yields

$$S^* = \mathcal{R}_0^{-1} \left(1 - \frac{\sigma\alpha\mathcal{R}_0^2}{\gamma\varepsilon}\right) + \mathcal{O}(\varepsilon^{-2}). \quad (47)$$

Therefore, (44) becomes

$$I^* = (1 - \mathcal{R}_0^{-1}) - \frac{\alpha}{\delta\varepsilon g} - (1 - \sigma)\frac{\alpha}{\gamma\varepsilon g} + \mathcal{O}(\varepsilon^{-2}). \quad (48)$$

[CM says: This didn't work as I expected, the ε term is going to upgrade the order of S^* . (UPDATE) This is easier than I thought, we just divide the polynomial (10) by ε^2 and repeat the process in that section. Much better than what I am trying here. We lose the clean closed form but it is much general and is already developed for the small ε part. Although we might want to motivate $R^* = \mathcal{O}(\varepsilon^{-1})$ (i.e., the first term of the series for R and C is 0) before presenting the scaled variables.]

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